

Project Data Page:



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WARM HP FLARE KO DRUM

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A170RHP10HN001\\WARM HP FLARE K.O. DRUM-7%.pvdb

Date of Analysis : Sep 17, 2026 3:11pm

PV Elite 28, January 2026



Customer : by:
item : date:
Item No : S/O:

Design Internal Pressure : Temp. :
Design External Pressure : Temp. :

Head Matl.:	Corr. All.:	JE :
Shell Matl.:	Corr. All.:	JE :
Flange Matl.:	Corr. All.:	JE :
Pipe Matl.:	Corr. All.:	JE :
Cone Matl.:	Corr. All.:	JE :

Radiographic Requirements :

Post Weld Heat Treat:

Wind Specification : V : Exp.: Imp :

Seismic : Zone : Cat.:

Results ...

Basic Flange Class: Rating : Ambient:

MAWP (Corroded) : limited by :
MAP (New & Cold): limited by :
Min. Design Metal Tmp: without Impacts
Charpy Impact Reqmnts.:

Stiffening Rings Required :

hydrotest Requirement :

Notes ...

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 3520.000 mm ID	
Vessel Design Length, Tangent to Tangent	10560.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	145 °C
Internal Design Pressure	0.350 MPa
External Design Temperature	145 °C
External Design Pressure	0.101 MPa
Maximum Allowable Working Pressure	0 MPa
External Max. Allowable Working Pressure	0 MPa
Shop Test Pressure	0.588 MPa
Field Test Pressure	0.588 MPa
Required Minimum Design Metal Temperature	0.0 °C
Warmest Computed Minimum Design Metal Temperature	-48.0 °C
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg		...	...		
Rings	SA-516 70	...	...	K02700	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-283 C	...	...	K02401	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
EH-1	0.358	0.10	1.000	1.5000	No	No
SHELL-1	0.358	0.10	2.000	1.5000	N/A	No
SHELL-2	0.358	0.10	2.000	1.5000	N/A	No
SHELL-3	0.358	0.10	1.000	1.5000	N/A	No
SHELL-4	0.358	0.10	1.000	1.5000	N/A	No
EH-2	0.358	0.10	1.000	1.5000	No	No

Liquid Level: 850.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000**Stiffener Ring Specifications:**

Elevation mm	Selected Type	Ring Description
5000.00	Bar 250.0 x 14.0	Ring R1 Fr20
7000.00	Bar 266.0 x 14.0	Ring R1 Fr40
9000.00	Bar 250.0 x 14.0	Ring R1 Fr50

**Element Types and Properties:**

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	14.0	12.0	6.1	...	1.00	1.00
Cylinder	3050.0	3000.0	22.0	22.0	6.1	...	1.00	1.00
Cylinder	6050.0	3000.0	22.0	22.0	6.1	...	1.00	1.00
Cylinder	8510.0	2460.0	12.0	12.0	6.1	...	1.00	1.00
Cylinder	10510.0	2000.0	12.0	12.0	6.1	...	1.00	1.00
Ellipse	10560.0	50.0	18.0	16.0	6.1	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	12	10.5813	0.1013	0.13543
20	30	22	14.8589	0.1013	0.29925
30	Ring	22	14.8589	0.1013	0.29925
Ring	40	22	10.4468	0.1013	0.67476
40	Ring	12	10.4153	0.1013	0.15545
Ring	50	12	10.4152	0.1013	0.15545
50	Ring	12	10.4152	0.1013	0.15545
Ring	60	12	10.1286	0.1013	0.16851
60	70	16	10.602	0.1013	0.25711

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	5293.33	14810.3	No Calc	No Calc
30	Ring	5293.33	14810.3	No Calc	No Calc
Ring	40	2000	14810.1	11490538	59112968
40	Ring	2000	3011.36	No Calc	No Calc
Ring	50	2000	3011.36	6182125	48622268
50	Ring	2000	3011.36	No Calc	No Calc
Ring	60	1853.33	3011.06	5960923	41345832
60	70	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-24''	WRC-297	0.892	Shell	Passed
N4-24''	WRC-297	0.892	Shell	Passed
N9-3''	WRC-297	0.580	Shell	Passed
N7-2''	WRC-297	0.661	Nozzle	Passed

Foundation/Support Summary**Saddle Parameters:**

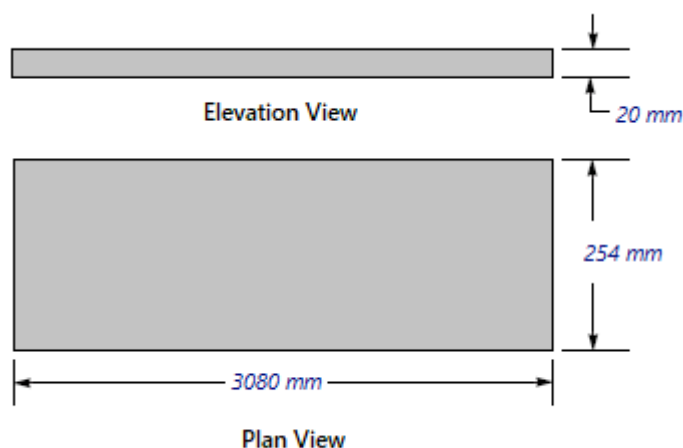
Saddle Width	228.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	2300.000 mm
Wear Pad Width	300.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	950.000 mm

Baseplate Length	3080.000 mm
------------------	-------------



Baseplate Thickness	20.000	mm
Baseplate Width	254.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	16.000	mm
Web Thickness	16.000	mm
Height of Center Web	498.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	394806.81	N
Maximum Transverse Saddle Shear Load	42994.89	N
Maximum Longitudinal Saddle Shear Load	52433.46	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	785379.06	N
Maximum Transverse Saddle Shear Load	14188.31	N
Maximum Longitudinal Saddle Shear Load	11900.55	N

Weights:

Fabricated - Bare w/o removable internals	27222.9	kgm
Shop Test - Fabricated + water (full)	141355.4	kgm
Shipping - Fab. + rem. intls.+ shipping app.	27222.9	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	37426.7	kgm
Empty - Fab. + intls. + details + wghts.	37426.7	kgm
Operating - Empty + operating liquid (no ca)	58185.5	kgm
Field Test - Empty weight + test medium (full)	151559.1	kgm

**Nozzle Calculation Summary (MPa & mm):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N11-2''	1	...	...	OK	4.92	OK	No Calc[*]
N10A-2''	1	...	...	OK	4.92	OK	No Calc[*]
N10B-2''	1	...	...	OK	4.92	OK	No Calc[*]
N3-24''	1	OK	...	OK	7.28	OK	Passed
M1-24	1	OK	...	...	...	OK	Passed
N4-24''	1	OK	...	OK	7.28	OK	Passed
N5-2''	2	...	...	OK	4.92	OK	No Calc[*]
N6-2''	1	...	...	OK	4.92	OK	No Calc[*]
N9-3''	1	...	...	OK	6.30	OK	No Calc[*]
N1-2''	1	...	...	OK	4.92	OK	No Calc[*]
N2-2''	1	...	...	OK	4.92	OK	No Calc[*]
N7-2''	1	...	...	OK	4.92	OK	No Calc[*]
M2-24''	1	OK	...	...	...	OK	Passed
N8A-2''	1	...	...	OK	4.92	OK	No Calc[*]
N8B-2''	1	...	...	OK	4.92	OK	No Calc[*]
N12A	1	...	...	OK	4.92	OK	No Calc[*]
N12B-2''	1	...	...	OK	4.92	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : N7-2''

Minimum MAWP Shells/Flanges : 0 MPa

Computed Vessel M.A.W.P. : 0 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N11-2''	0.000	0.000	71.800
10	N10A-2''	0.000	90.000	71.800
10	N10B-2''	0.000	270.000	71.800
20	N3-24''	2550.000	90.000	1177.200
20	M1-24	1050.000	90.000	1177.200
30	N4-24''	4050.000	90.000	1177.200
30	N5-2''	5450.000	90.000	91.800
40	N6-2''	6250.001	90.000	71.800
40	N9-3''	7250.001	270.000	144.862
40	N1-2''	6800.000	90.000	71.800
40	N2-2''	7500.000	90.000	71.800
50	N7-2''	9110.000	90.000	71.800
60	M2-24''	0.000	0.000	1169.200
60	N8A-2''	0.000	90.000	79.800
60	N8B-2''	0.000	270.000	79.800
60	N12A	0.000	90.000	79.800
60	N12B-2''	0.000	270.000	79.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

WARM HP FLARE KO DRUM

PV Elite 28

Licensee: SL User



FileName : A170RHP10HN001_WARM HP FLARE K.O. DRUM-7%

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Nozzle Summary:

Step: 40 3:11pm Sep 17, 2026



HEXAGON

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N11-2''	2.000 in	Actual	LWN	2.000	12.000	...	...	212.10	150
N10A-2''	2.000 in	Actual	LWN	2.000	12.000	...	...	203.65	150
N10B-2''	2.000 in	Actual	LWN	2.000	12.000	...	...	203.65	150
N5-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	894.18	150
N6-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	900.18	150
N1-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	884.18	150
N2-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	884.18	150
N7-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	900.18	300
N8A-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	225.38	150
N8B-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	225.38	150
N12A	2.000 in	Actual	LWN	2.000	10.000	...	...	229.03	150
N12B-2''	2.000 in	Actual	LWN	2.000	10.000	...	...	229.03	150
N9-3''	3.000 in	160	WNF	3.500	11.125	...	...	212.56	300
N3-24''	24.000 in	Actual	WNF	24.000	12.000	...	...	920.59	150
M1-24	24.000 in	Actual	WNF	24.000	12.000	...	...	920.59	150
N4-24''	24.000 in	Actual	WNF	24.000	12.000	...	...	920.59	150
M2-24''	24.000 in	Actual	WNF	24.000	14.000	...	...	330.64	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N11-2''	SA-106 B	12.000	10.000	...	...	...
N10A-2''	SA-106 B	12.000	10.000	...	...	...
N10B-2''	SA-106 B	12.000	10.000	...	...	...
N5-2''	SA-106 B	12.000	10.000	...	...	...
N6-2''	SA-106 B	12.000	10.000	...	...	...
N1-2''	SA-106 B	12.000	10.000	...	...	...
N2-2''	SA-106 B	12.000	10.000	...	...	...
N7-2''	SA-106 B	12.000	10.000	...	...	...
N8A-2''	SA-106 B	14.000	10.000	...	...	...
N8B-2''	SA-106 B	14.000	10.000	...	...	...
N12A	SA-106 B	14.000	10.000	...	...	...
N12B-2''	SA-106 B	14.000	10.000	...	...	...
N9-3''	SA-106 B	12.000	10.000	...	...	...
N3-24''	SA-516 70	12.000	10.000	...	...	...
M1-24	SA-516 70	12.000	10.000	...	...	...
N4-24''	SA-516 70	12.000	10.000	...	...	...
M2-24''	SA-516 70	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N11-2''	...	0.0	200.00	0.00	EH-1
N10A-2''	...	90.0	182.47	0.00	EH-1



N10B-2''	...	270.0	182.47	0.00	EH-1
N5-2''	5450.000	90.0	872.00	0.00	SHELL-2
N6-2''	6250.001	90.0	888.00	0.00	SHELL-3
N1-2''	6800.000	90.0	872.00	0.00	SHELL-3
N2-2''	7500.000	90.0	872.00	0.00	SHELL-3
N7-2''	9110.000	90.0	888.00	0.00	SHELL-4
N8A-2''	...	90.0	200.00	0.00	EH-2
N8B-2''	...	270.0	200.00	0.00	EH-2
N12A	...	90.0	200.00	0.00	EH-2
N12B-2''	...	270.0	200.00	0.00	EH-2
N9-3''	7250.000	270.0	200.00	0.00	SHELL-3
N3-24''	2550.000	90.0	872.00	0.00	SHELL-1
M1-24	1050.000	90.0	872.00	0.00	SHELL-1
N4-24''	4050.000	90.0	872.00	0.00	SHELL-2
M2-24''	...	0.0	300.00	0.00	EH-2

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 3520.0mm ID X 50.0mm	SA-516 70
2	CYLINDER: 22.0mm THK X 3520.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 3520.0mm ID X 2460.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 3520.0mm ID X 2000.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm THK X 3520.0mm ID X 50.0mm	SA-516 70
2	INSULATION: 930mm X 100mm THK	PP
2	SADDLE: 228.0mm X 120 DEG	SA-283 C
1	PLATFORM: 1000mm X 3563mm WIDE	
1	LADDER: 4000.0mm	...
2	INSULATION: 3000mm X 100mm THK	PP
1	BAR RING STIFFENER: 3564.0mm ID X 4064.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 2460mm X 100mm THK	PP
1	BAR RING STIFFENER: 3544.0mm ID X 4076.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 2000mm X 100mm THK	PP
1	BAR RING STIFFENER: 3560.0mm ID X 4060.0mm OD X 14.0mm	SA-516 70
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
EH-1	[10]	B	-23	-48	-29	0.559	12.000	1.00	No
EH-1	[7]	B	-18	-48	-29	0.471	14.000	1.00	No
SHELL-1	[8]	B	-21	-104	-29	0.287	22.000	1.00	No
SHELL-2	[8]	B	-21	-104	-29	0.287	22.000	1.00	No
SHELL-3	[8]	B	-23	-48	-29	0.561	12.000	1.00	No
SHELL-4	[8]	B	-23	-48	-29	0.561	12.000	1.00	No
EH-2	[10]	B	-14	-48	-29	0.405	16.000	1.00	No
EH-2	[7]	B	-11	-48	-29	0.357	18.000	1.00	No
Ring R1 Fr20	[23]	B	-35	-104		0.287	14.000	1.00	No
Ring R1 Fr40	[23]	B	-23	-48	-29	0.561	12.000	1.00	No
Ring R1 Fr50	[23]	B	-23	-48	-29	0.561	12.000	1.00	No
Left Saddle	[24]	B	-46	-104		0.287	10.000	1.00	No
RIGHT SADDLE	[24]	B	-29	-48	-29	0.561	10.000	1.00	No
N11-2''	[1]	B	-23	-48	-29	0.494	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N10A-2''	[1]	B	-23	-48	-29	0.494	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N10B-2''	[1]	B	-23	-48	-29	0.495	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-24''	[1]	B	-19	-104	-29	0.095	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24	[1]	B	-19	-104	-29	0.095	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-24''	[1]	B	-19	-104	-29	0.095	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-2''	[1]	B	-46	-104		0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N9-3''	[1]	B	-23	-48	-29	0.561	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
N2-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N7-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M2-24''	[1]	B	-14	-48	-29	0.358	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N8A-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N8B-2''	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N12A	[1]	B	-29	-104	-29	0.011	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N12B-2''	[1]	B	-29	-104	-29	0.012	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -11 -48

Required Minimum Design Metal Temperature 0.0 °C

Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.



- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (SIASME):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

WARM HP FLARE KO DRUM

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	145.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	0.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	0 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm
Distance Upwind of Crest	Lh 0 mm
Distance from Crest to the Vessel	x 0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes



Consider Longitudinal Stress in MDMT Calculations

Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	EH-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	3520 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	145 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	145 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	232.6 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009984 kg/cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-880 mm
Height/Length of Insulation	930 mm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	N11-2''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	53.083 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	10
Detail Type	Nozzle
Detail ID	N10A-2''
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.



Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	51.134 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	10
Detail Type	Nozzle
Detail ID	N10B-2''
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	51.134 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	SHELL-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3520 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	145 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	145 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	950 mm
Width of Saddle	228 mm
Height of Saddle at Bottom	2300 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	300 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	20
Detail Type	Platform
Detail ID	PLATFORM
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	15841300 mm ²



Platform Weight	64732	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	3564	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.3	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	10560	mm

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	850	mm
Liquid Density	0.0009984	kg/cm ³

Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	100	mm
Density	0.0002	kg/cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N3-24''	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	3974.1	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	M1-24	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5624.3	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	20	
Detail Type	Weight	
Detail ID	cleat weight	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	2451.5	N
Offset from Element Centerline	0	mm

Element From Node	20	
Detail Type	Weight	
Detail ID	LL-1	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	4118.5	N
Offset from Element Centerline	1600	mm

Element From Node	20	
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Detail Type	Weight
Detail ID	M1 davit
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	490.31 N
Offset from Element Centerline	1860 mm

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	SHELL-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3520 mm
Element Thickness	22 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	145 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	145 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm³

Element From Node	30
Detail Type	Ring
Detail ID	Ring R1 Fr20
Dist. from "FROM" Node / Offset dist	1950 mm
Inside Diameter of Ring	3564 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	4064 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Nozzle
Detail ID	N4-24''
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	3623.1 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	30
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Detail Type	Nozzle
Detail ID	N5-2''
Dist. from "FROM" Node / Offset dist	2400 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	117.53 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	SHELL-3
Distance "FROM" to "TO"	2460 mm
Inside Diameter	3520 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	145 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	145 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2460 mm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm³

Element From Node	40
Detail Type	Ring
Detail ID	Ring R1 Fr40
Dist. from "FROM" Node / Offset dist	950 mm
Inside Diameter of Ring	3544 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	4076 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element From Node	40
Detail Type	Nozzle
Detail ID	N6-2''
Dist. from "FROM" Node / Offset dist	200 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.53	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N9-3''	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	188.4	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1-2''	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	115.97	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-2''	
Dist. from "FROM" Node / Offset dist	1450	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	115.97	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Weight	
Detail ID	VB	
Dist. from "FROM" Node / Offset dist	1200	mm
Miscellaneous Weight	98.062	N
Offset from Element Centerline	1560	mm

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	SHELL-4	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	3520	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	145	°C
Design External Pressure	0.1013	MPa
Design Temperature External Pressure	145	°C
Effective Diameter Multiplier	1.2	



Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	RIGHT SADDLE
Dist. from "FROM" Node / Offset dist	1050 mm
Width of Saddle	228 mm
Height of Saddle at Bottom	2300 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	300 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.60000002
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009984 kg/cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2000 mm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm³

Element From Node	50
Detail Type	Ring
Detail ID	Ring R1 Fr50
Dist. from "FROM" Node / Offset dist	490 mm
Inside Diameter of Ring	3560 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	4060 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element From Node	50
Detail Type	Nozzle
Detail ID	N7-2''
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	128.11 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	50
Detail Type	Weight
Detail ID	LL-2
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	4118.5 N
Offset from Element Centerline	1600 mm



Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	EH-2
Distance "FROM" to "TO"	50 mm
Inside Diameter	3520 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	145 °C
Design External Pressure	0.1013 MPa
Design Temperature External Pressure	145 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	850 mm
Liquid Density	0.0009984 kg/cm³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	930 mm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm³
Element From Node	60
Detail Type	Nozzle
Detail ID	M2-24''
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	4392.3 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70
Element From Node	60
Detail Type	Nozzle
Detail ID	N8A-2''
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	50.709 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	60
Detail Type	Nozzle
Detail ID	N8B-2''
Dist. from "FROM" Node / Offset dist	1000 mm



Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.709	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N12A	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.711	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N12B-2''	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.711	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Weight	
Detail ID	M2-davit	
Dist. from "FROM" Node / Offset dist	880	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	EH-1	50	...	50	...
	SHELL-1	3050	...	3000	...
	SHELL-2	6050	...	3000	...
	SHELL-3	8510	...	2460	...
	SHELL-4	10510	...	2000	...
	EH-2	10560	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	EH-1	0.3583	14	1.5	3520	138	1
	SHELL-1	0.3583	22	1.5	3520	138	2
	SHELL-2	0.3583	22	1.5	3520	138	3
	SHELL-3	0.3583	12	1.5	3520	138	4
	SHELL-4	0.3583	12	1.5	3520	138	5
	EH-2	0.3583	18	1.5	3520	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	EH-1	0.35	0.81468	0.94025	12	6.0702	1
	SHELL-1	0.35	1.58651	1.71213	22	6.08134	2
	SHELL-2	0.35	1.58651	1.71213	22	6.08134	3
	SHELL-3	0.35	0.8113	0.93706	12	6.08134	4
	SHELL-4	0.35	0.8113	0.93706	12	6.08134	5
	EH-2	0.35	1.12795	1.25338	16	6.0702	6

Minimum

0

1

Warning: A standard flange is governing the MAWP!

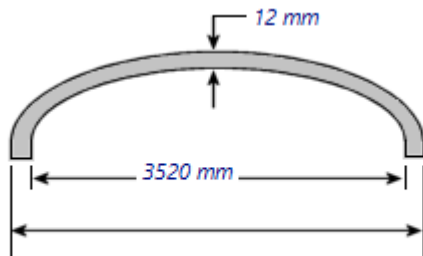
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-1

Elliptical Head Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		145 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3520.000 mm	Actual Stress		60.1 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		14202853.0 mm²	SG of Contents		1.000
Element Volume		6196.8 ltr	Weight of Contents		8913.2 N
Empty Weight		16495.3 N	Operating Weight		25408.5 N
			Comp. All. Ope. Stress		88.9 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 44.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.358 \cdot 3523.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.358)$$

$$= 4.5702 + 1.5000 = 6.0702 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.5) / (0.999 \cdot 3523.0 + 0.2 \cdot 10.5)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 3520.0 + 0.2 \cdot 12.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.358 \cdot (0.999 \cdot 3523.0 + 0.2 \cdot 10.5)) / (2 \cdot 1.0 \cdot 10.5)$$

$$= 60.085 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 - 0.6 \cdot 0.358) + 1.5$$

$$= 6.081 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.5) / (1761.4999 + 0.6 \cdot 12.5)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3523.0 / (2 \cdot 881.5)))^2 / 6$$

$$= 0.998866$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.734 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.0, tr = 5.871, c = 1.5 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c)$ = 0.559, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 5.887, c = 1.5 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c)$ = 0.471, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.559	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 44.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.000 mm

EXTERNAL PRESSURE RESULTS for: EH-1

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.500	3544.00	337.52	0.0004115	41.14

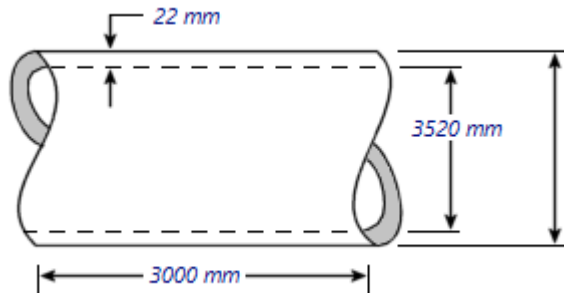
$$MAEP = B / (K \cdot Do / t) = 41.1401 / (0.9 \cdot 337.5238) = 0.1354 \text{ MPa}$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
9.081	3544.00	390.25	0.0003559	35.58

$$MAEP = B / (K0 * Do / t) = 35.5816 / (0.9 * 390.2514) = 0.1013 \text{ MPa}$$

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**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: SHELL-1**

Cylindrical Shell Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		145 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3520.000 mm	Actual Stress		31.0 MPa
Spec. Min. Thk t,e, or sa		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		33589908.0 mm²	SG of Contents		1.000
Element Volume		29199.4 ltr	Weight of Contents		53269.9 N
Empty Weight		55816.6 N	Operating Weight		109086.5 N
			Comp. All. Ope. Stress		114.8 MPa
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 0.358)$$

$$= 4.5813 + 1.5000 = 6.0813 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1761.4999 + 0.6 \cdot 20.5)$$

$$= 2 - 0.01 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1760.0 + 0.6 \cdot 22.0)$$

$$= 2 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.358 \cdot (1761.4999 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 31.007 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.621 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 22.0$, $t_r = 5.887$, $c = 1.5$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.287$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	22.000 mm	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.287	Temperature Difference	83 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: SHELL-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	5293.33	173.85	1.4852	0.0003903	39.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.0199)/(3*173.8537) = 0.2993 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.359	3564.00	5293.33	266.79	1.4852	0.0002028	20.27

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.2705)/(3*266.7885) = 0.1013 \text{ MPa}$$

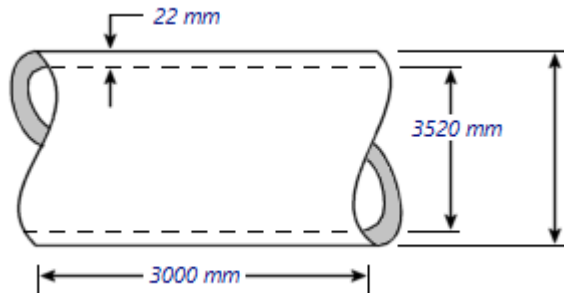
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	14810.33	173.85	4.1555	0.0001322	13.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.2154)/(3*173.8537) = 0.1014 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		145 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3520.000 mm	Actual Stress		31.0 MPa
Spec. Min. Thk t,e, or sa		22.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		33589908.0 mm²	SG of Contents		1.000
Element Volume		29199.4 ltr	Weight of Contents		53269.9 N
Empty Weight		55816.6 N	Operating Weight		109086.5 N
			Comp. All. Ope. Stress		114.8 MPa
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 0.358)$$

$$= 4.5813 + 1.5000 = 6.0813 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.5) / (1761.4999 + 0.6 \cdot 20.5)$$

$$= 2 - 0.01 = 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 22.0) / (1760.0 + 0.6 \cdot 22.0)$$

$$= 2 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.358 \cdot (1761.4999 + 0.6 \cdot 20.5)) / (1.0 \cdot 20.5)$$

$$= 31.007 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.621 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 22.0, tr = 5.887, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.287, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	22.000 mm	Basic MDMT	-21 °C
Stress/Thk. Ratio	0.287	Temperature Difference	83 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cylindrical Shell From 30 to Ring R1 Fr20 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	5293.33	173.85	1.4852	0.0003903	39.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.0199)/(3*173.8537) = 0.2993 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.359	3564.00	5293.33	266.79	1.4852	0.0002028	20.27

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.2705)/(3*266.7885) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	14810.33	173.85	4.1555	0.0001322	13.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.2154)/(3*173.8537) = 0.1014 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr20 to 40 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	2000.00	173.85	0.5612	0.0010928	87.98

$$MAEP = (4*B)/(3*(Do/t)) = (4*87.9819)/(3*173.8537) = 0.6748 \text{ MPa}$$

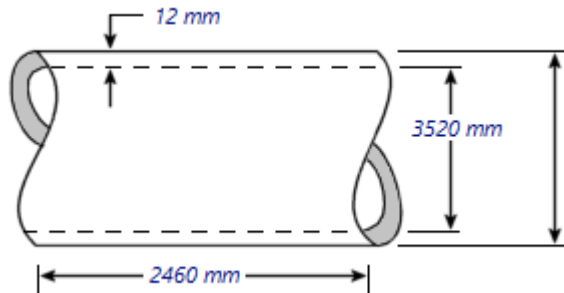
Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.947	3564.00	2000.00	398.35	0.5612	0.0003027	30.27



$$MAEP = (4*B)/(3*(Do/t)) = (4*30.2655)/(3*398.3532) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
20.500	3564.00	14810.15	173.85	4.1555	0.0001322	13.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.2155)/(3*173.8537) = 0.1014 \text{ MPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: SHELL-3**

Cylindrical Shell Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		145 °C
Length		2460.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3520.000 mm	Actual Stress		60.3 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		27389160.0 mm²	SG of Contents		1.000
Element Volume		23943.5 ltr	Weight of Contents		43681.3 N
Empty Weight		24894.8 N	Operating Weight		68576.1 N
			Comp. All. Ope. Stress		88.9 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 0.358) \\
 &= 4.5813 + 1.5000 = 6.0813 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 10.5) / (1761.4999 + 0.6 \cdot 10.5) \\
 &= 1 - 0.01 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (1760.0 + 0.6 \cdot 12.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.358 \cdot (1761.4999 + 0.6 \cdot 10.5)) / (1.0 \cdot 10.5) \\
 &= 60.332 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.340 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 5.887$, $c = 1.5$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.561$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.561	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: SHELL-3

Cylindrical Shell From 40 to Ring R1 Fr40 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	2000.00	337.52	0.5643	0.0003936	39.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.352)/(3*337.5238) = 0.1555 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.915	3544.00	2000.00	397.52	0.5643	0.0003021	30.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.2025)/(3*397.521) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	3011.36	337.52	0.8497	0.0002566	25.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6582)/(3*337.5238) = 0.1014 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr40 to 50 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	2000.00	337.52	0.5643	0.0003936	39.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.3521)/(3*337.5238) = 0.1555 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.915	3544.00	2000.00	397.52	0.5643	0.0003021	30.20



$$MAEP = (4*B)/(3*(Do/t)) = (4*30.2025)/(3*397.5211) = 0.1013 \text{ MPa}$$

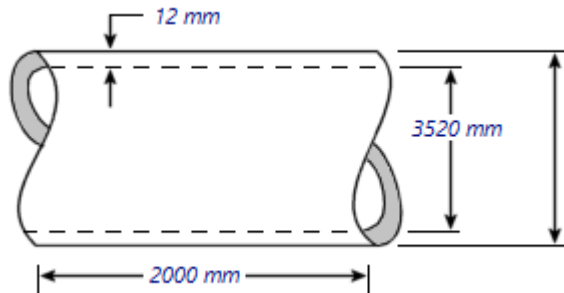
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	3011.36	337.52	0.8497	0.0002566	25.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6582)/(3*337.5238) = 0.1014 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-4

Cylindrical Shell Design Information:			
Design Pressure P	0.358 MPa	Int. Design Temperature	145 °C
Material	SA-516 70	Ext. Design Temperature	145 °C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	3520.000 mm	Actual Stress	60.3 MPa
Spec. Min. Thk t, e, or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	22267608.0 mm ²	SG of Contents	1.000
Element Volume	19466.3 ltr	Weight of Contents	35513.3 N
Empty Weight	20239.8 N	Operating Weight	55753.1 N
		Comp. All. Ope. Stress	88.9 MPa
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 0.358) \\
 &= 4.5813 + 1.5000 = 6.0813 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 10.5) / (1761.4999 + 0.6 \cdot 10.5) \\
 &= 1 - 0.01 = 1 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (1760.0 + 0.6 \cdot 12.0) \\
 &= 1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.358 \cdot (1761.4999 + 0.6 \cdot 10.5)) / (1.0 \cdot 10.5) \\
 &= 60.332 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.340 %

Minimum Design Metal Temperature Results:



Govern. thk, $t_g = 12.0$, $t_r = 5.887$, $c = 1.5$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.561$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.561	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: SHELL-4

Cylindrical Shell From 50 to Ring R1 Fr50 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	2000.00	337.52	0.5643	0.0003936	39.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.3521)/(3*337.5238) = 0.1555 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.915	3544.00	2000.00	397.52	0.5643	0.0003021	30.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*30.2025)/(3*397.5211) = 0.1013 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	3011.36	337.52	0.8497	0.0002566	25.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6582)/(3*337.5238) = 0.1014 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr50 to 60 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	1853.33	337.52	0.5229	0.0004267	42.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*42.6561)/(3*337.5238) = 0.1685 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.629	3544.00	1853.33	410.73	0.5229	0.0003121	31.21



$$MAEP = (4*B)/(3*(Do/t)) = (4*31.2062)/(3*410.7281) = 0.1013 \text{ MPa}$$

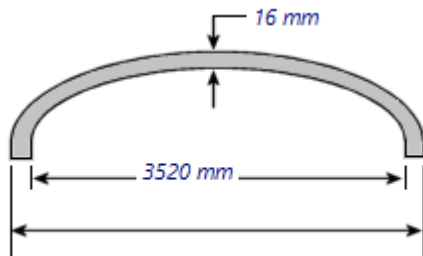
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	3544.00	3011.06	337.52	0.8496	0.0002567	25.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6609)/(3*337.5238) = 0.1014 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: EH-2

Elliptical Head Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		145 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3520.000 mm	Actual Stress		43.5 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		14265717.0 mm ²	SG of Contents		1.000
Element Volume		6196.8 ltr	Weight of Contents		8913.2 N
Empty Weight		21255.4 N	Operating Weight		30168.6 N
			Comp. All. Ope. Stress		88.9 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 44.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.358 \cdot 3523.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.358)$$

$$= 4.5702 + 1.5000 = 6.0702 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.5) / (0.999 \cdot 3523.0 + 0.2 \cdot 14.5)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 16.0) / (1.0 \cdot 3520.0 + 0.2 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.358 \cdot (0.999 \cdot 3523.0 + 0.2 \cdot 14.5)) / (2 \cdot 1.0 \cdot 14.5)$$

$$= 43.519 \text{ MPa}$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.358 \cdot 1761.4999) / (138.0 \cdot 1.0 - 0.6 \cdot 0.358) + 1.5$$

$$= 6.081 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.008 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.5) / (1761.4999 + 0.6 \cdot 16.5)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (3523.0 / (2 \cdot 881.5))^2) / 6$$

$$= 0.998866$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.223 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 16.0, tr = 5.871, c = 1.5 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c)$ = 0.405, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 18.0, tr = 5.887, c = 1.5 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c)$ = 0.357, Temp. Reduction = 73 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.460 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.357	Temperature Difference	38 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 44.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 22.000 mm

EXTERNAL PRESSURE RESULTS for: EH-2

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 145 °C

Elastic Modulus from Chart: CS-2 at 145 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
14.500	3552.00	244.97	0.0005670	56.68

$$MAEP = B / (K0 \cdot Do / t) = 56.6846 / (0.9 \cdot 244.9655) = 0.2571 \text{ MPa}$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
9.102	3552.00	390.25	0.0003559	35.58

$$MAEP = B / (K0 * Do / t) = 35.5822 / (0.9 * 390.2452) = 0.1013 \text{ MPa}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.588 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.184 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.497 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	0.452 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 0.452 • 1.0

= 0.588 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	91.3	234.0	0.390	0.62
SHELL-1	50.1	234.0	0.214	0.62
SHELL-2	50.1	234.0	0.214	0.62
SHELL-3	91.6	234.0	0.391	0.62
SHELL-4	91.6	234.0	0.391	0.62
EH-2	68.5	234.0	0.293	0.62

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N11-2''	Nozzle	117.98	117.98	1.000
N10A-2''	Nozzle	117.98	117.98	1.000
N10B-2''	Nozzle	117.98	117.98	1.000
N3-24''	Nozzle	137.98	137.98	1.000
M1-24	Nozzle	137.98	137.98	1.000
N4-24''	Nozzle	137.98	137.98	1.000
N5-2''	Nozzle	117.98	117.98	1.000
N6-2''	Nozzle	117.98	117.98	1.000
N9-3''	Nozzle	117.98	117.98	1.000
N1-2''	Nozzle	117.98	117.98	1.000
N2-2''	Nozzle	117.98	117.98	1.000
N7-2''	Nozzle	117.98	117.98	1.000
M2-24''	Nozzle	137.98	137.98	1.000
N8A-2''	Nozzle	117.98	117.98	1.000
N8B-2''	Nozzle	117.98	117.98	1.000
N12A	Nozzle	117.98	117.98	1.000
N12B-2''	Nozzle	117.98	117.98	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	138.00	1.000
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr50	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
SHELL-3	138.00	138.00	1.000
SHELL-4	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N11-2''	1.07	153.38	0.007
N10A-2''	1.07	153.38	0.007
N10B-2''	1.07	153.38	0.007
N3-24''	15.55	179.37	0.087
M1-24	15.55	179.37	0.087
N4-24''	15.55	179.37	0.087
N5-2''	1.33	153.38	0.009
N6-2''	1.33	153.38	0.009
N9-3''	2.59	153.38	0.017
N1-2''	1.33	153.38	0.009
N2-2''	1.33	153.38	0.009
N7-2''	1.33	153.38	0.009
M2-24''	13.29	179.37	0.074
N8A-2''	1.33	153.38	0.009
N8B-2''	1.33	153.38	0.009
N12A	1.33	153.38	0.009
N12B-2''	1.33	153.38	0.009

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.588 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.184 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.497 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	0.452 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 0.452 • 1.0

= 0.588 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	104.3	234.0	0.446	0.62
SHELL-1	53.8	234.0	0.230	0.62
SHELL-2	53.8	234.0	0.230	0.62
SHELL-3	104.7	234.0	0.448	0.62
SHELL-4	104.7	234.0	0.448	0.62
EH-2	75.5	234.0	0.323	0.62

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N11-2''	Nozzle	117.98	117.98	1.000
N10A-2''	Nozzle	117.98	117.98	1.000
N10B-2''	Nozzle	117.98	117.98	1.000
N3-24''	Nozzle	137.98	137.98	1.000
M1-24	Nozzle	137.98	137.98	1.000
N4-24''	Nozzle	137.98	137.98	1.000
N5-2''	Nozzle	117.98	117.98	1.000
N6-2''	Nozzle	117.98	117.98	1.000
N9-3''	Nozzle	117.98	117.98	1.000
N1-2''	Nozzle	117.98	117.98	1.000
N2-2''	Nozzle	117.98	117.98	1.000
N7-2''	Nozzle	117.98	117.98	1.000
M2-24''	Nozzle	137.98	137.98	1.000
N8A-2''	Nozzle	117.98	117.98	1.000
N8B-2''	Nozzle	117.98	117.98	1.000
N12A	Nozzle	117.98	117.98	1.000
N12B-2''	Nozzle	117.98	117.98	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	138.00	1.000
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr50	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
EH-1	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
SHELL-3	138.00	138.00	1.000
SHELL-4	138.00	138.00	1.000
EH-2	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N11-2''	1.07	153.38	0.007
N10A-2''	1.07	153.38	0.007
N10B-2''	1.07	153.38	0.007
N3-24''	15.55	179.37	0.087
M1-24	15.55	179.37	0.087
N4-24''	15.55	179.37	0.087
N5-2''	1.33	153.38	0.009
N6-2''	1.33	153.38	0.009
N9-3''	2.59	153.38	0.017
N1-2''	1.33	153.38	0.009
N2-2''	1.33	153.38	0.009
N7-2''	1.33	153.38	0.009
M2-24''	13.29	179.37	0.074
N8A-2''	1.33	153.38	0.009
N8B-2''	1.33	153.38	0.009
N12A	1.33	153.38	0.009
N12B-2''	1.33	153.38	0.009

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

MDMT Calculations for: Ring R1 Fr20, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 14.0$, $c = 1.5$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.287$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

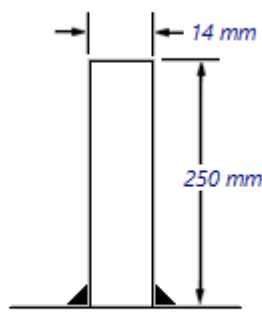
Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -35 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Stiffening Ring Calculations for: Ring R1 Fr20, SA-516 70 , Bar Ring: 250.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 297 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	6095.261	10.2500	62476.422
Ring :	3500.001	145.5000	509250.188
Total:	9595.262		571726.625

Centroid of Ring plus Shell: 60 mm

	Inertia	Distance	A*Dist ²
Shell:	213460.891	49.3343	14835065.000
Ring :	18229162.000	-85.9158	25835320.000
Total:	18442622.000		40670384.000

Available Moment of Inertia, Ring + Shell: 0.591E+08 mm⁴

Required Stress in Ring plus Shell Breq 12.62 MPa

Required Strain in Ring plus Shell Areq 0.0001260

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3564.0^2 \cdot 3646.6665(20.5 + 3500.0007/3646.6665)0.000126) / 10.9$$

$$= 11490538 \text{ mm}^4$$

MDMT Calculations for: Ring R1 Fr40, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.887$, $c = 1.5$ mm, $E^* = 1.0$



Thickness Ratio = $tr * E^* / (tg - c) = 0.561$, Temp. Reduction = 26 °C

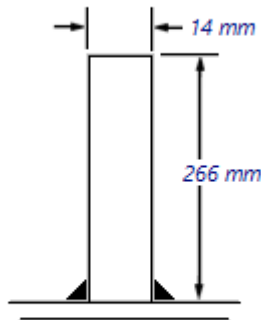
Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr40, SA-516 70 , Bar Ring: 266.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 212 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2228.043	5.2500	11697.225
Ring :	3724.001	143.5000	534394.188
Total:	5952.043		546091.438

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist ²
Shell:	20470.123	86.4986	16670220.000
Ring :	21957934.000	-51.7515	9973668.000
Total:	21978404.000		26643888.000

Available Moment of Inertia, Ring + Shell: 0.486E+08 mm⁴

Required Stress in Ring plus Shell Breq 21.78 MPa

Required Strain in Ring plus Shell Areq 0.0002170

Required Moment of Inertia, Ring plus Shell:

$$= (Do^2 \cdot Ls(t + As/Ls)A) / 10.9$$

$$= (3544.0^2 \cdot 2000.0(10.5 + 3724.0007/2000.0)0.000217) / 10.9$$

$$= 6182125 \text{ mm}^4$$

MDMT Calculations for: Ring R1 Fr50, Shell to Ring Junction, Curve: B

Govern. thk, $tg = 12.0$, $tr = 5.887$, $c = 1.5$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (tg - c) = 0.561$, Temp. Reduction = 26 °C

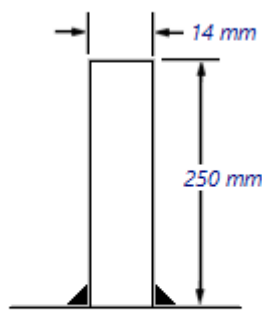
Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr50, SA-516 70 , Bar Ring: 250.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 212 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2228.043	5.2500	11697.225
Ring :	3500.001	135.5000	474250.188
Total:	5728.043		485947.406

Centroid of Ring plus Shell: 85 mm

	Inertia	Distance	A*Dist ²
Shell:	20470.123	79.5865	14112465.000
Ring :	18229162.000	-50.6635	8983759.000
Total:	18249632.000		23096224.000

Available Moment of Inertia, Ring + Shell: 0.413E+08 mm⁴

Required Stress in Ring plus Shell Breq 21.86 MPa
Required Strain in Ring plus Shell Areq 0.0002180

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3544.0^2 \cdot 1926.6666(10.5 + 3500.0007/1926.6666)0.000218) / 10.9$$

$$= 5960923 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	3544	10.5	0.00041149	41.1401
20	30	5293.33	3564	20.5	0.00039029	39.0199
30	Ring	5293.33	3564	20.5	0.00039029	39.0199
Ring	40	2000	3564	20.5	0.0010928	87.9819
40	Ring	2000	3544	10.5	0.00039361	39.352
Ring	50	2000	3544	10.5	0.00039361	39.3521
50	Ring	2000	3544	10.5	0.00039361	39.3521
Ring	60	1853.33	3544	10.5	0.00042666	42.6561
60	70	No Calc	3552	14.5	0.00056697	56.6846

External Pressure Calculations:

From	To	External Actual T.	External Required T.	External Design Pressure	External M.A.W.P.
------	----	--------------------	----------------------	--------------------------	-------------------



		mm	mm	MPa	MPa
10	20	12	10.5813	0.1013	0.13543
20	30	22	14.8589	0.1013	0.29925
30	Ring	22	14.8589	0.1013	0.29925
Ring	40	22	10.4468	0.1013	0.67476
40	Ring	12	10.4153	0.1013	0.15545
Ring	50	12	10.4152	0.1013	0.15545
50	Ring	12	10.4152	0.1013	0.15545
Ring	60	12	10.1286	0.1013	0.16851
60	70	16	10.602	0.1013	0.25711

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	5293.33	14810.3	No Calc	No Calc
30	Ring	5293.33	14810.3	No Calc	No Calc
Ring	40	2000	14810.1	11490538	59112968
40	Ring	2000	3011.36	No Calc	No Calc
Ring	50	2000	3011.36	6182125	48622268
50	Ring	2000	3011.36	No Calc	No Calc
Ring	60	1853.33	3011.06	5960923	41345832
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1682.16	6196.77	1501.93	6212.21	117.751
20	30	5692.09	29199.4	5306.27	29249.2	398.446
30	40	5692.09	29199.4	5306.27	29249.2	398.446
40	50	2538.73	23943.5	2222.37	23984.4	177.711
50	60	2064.02	19466.3	1806.8	19499.5	144.482
60	70	2167.59	6196.77	1986.96	6212.21	151.731
Total		19836	114202.18	18130	114406.66	1388

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	908.957	-293.333	1335	...	Liquid: 10
10	Insl	373.639	-415	...	...	Ins: 10
10	Nozl	5.7922	-880	...	...	N11-2''
10	Nozl	5.57962	-724.155	1000	...	N10A-2''
10	Nozl	5.57962	-724.155	-1000	...	N10B-2''
20	Sadl	843.147	950	2014	...	Left Saddle
20	Plat	7063.34	5280	1858.2	...	PLATFORM
20	Liqd	5432.38	1500	1335	...	Liquid 20
20	Insl	691.456	1500	...	...	Ins: 10
20	Nozl	433.64	2500	2064.8	...	N3-24''
20	Nozl	613.709	1000	2064.8	...	M1-24
20	Wght	250	1500	...	...	cleat weight
20	Wght	420	1000	1600	...	LL-1
20	Wght	50.0009	1000	1860	...	M1 davit
30	Liqd	5432.38	1500	1335	...	Liquid 30
30	Insl	691.456	1500	...	...	Ins: 10
30	Ring	347.768	1950	...	...	Ring R1 Fr20
30	Nozl	395.336	1000	2064.8	...	N4-24''
30	Nozl	12.8248	2400	1785.4	...	N5-2''
40	Liqd	4454.55	1230	1335	...	Liquid 40
40	Insl	563.9	1230	...	...	Ins: 10
40	Ring	369.637	950	...	...	Ring R1 Fr40
40	Nozl	12.8248	200.001	1785.4	...	N6-2''
40	Nozl	20.5579	1200	-1804.45	...	N9-3''
40	Nozl	12.6547	750	1785.4	...	N1-2''
40	Nozl	12.6547	1450	1785.4	...	N2-2''
40	Wght	10.0002	1200	-1560	...	VB
50	Sadl	843.181	1050	2019	...	RIGHT SADDLE
50	Liqd	3621.59	1000	1335	...	Liquid 50
50	Insl	458.455	1000	...	...	Ins: 10
50	Ring	347.404	490	...	...	Ring R1 Fr50
50	Nozl	13.9791	600	1785.4	...	N7-2''
50	Wght	420	1000	1600	...	LL-2
60	Liqd	908.957	343.333	1335	...	Liquid 60
60	Insl	361.48	465	...	...	Ins: 10
60	Nozl	479.277	930	...	...	M2-24''
60	Nozl	5.53316	774.155	1000	...	N8A-2''
60	Nozl	5.53316	774.155	-1000	...	N8B-2''
60	Nozl	5.53344	643.211	1300	...	N12A
60	Nozl	5.53344	643.211	-1300	...	N12B-2''
60	Wght	50	643.211	...	...	M2-davit

Total Weight of Each Detail Type:

Saddles 1686.3
Platforms 7063.3



Liquid	20758.8
Insulation	3140.4
Stiffeners	1064.8
Nozzles	2046.5
Weights	1200.0

Sum of the Detail Weights 36960.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	21225.2	21225.2	21225.2	21225.2	21225.2	21225.2
Saddles	1686.3	1686.3	1686.3	1686.3	1686.3	1686.3
Stif. Rings	1064.8	1064.8	1064.8	1064.8	1064.8	1064.8
Nozzles	2046.5	2046.5	2046.5	2046.5	2046.5	2046.5
Wld Weights	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0
Platforms	...	...	...	7063.3	7063.3	7063.3
Insulation	...	...	...	3140.4	3140.4	3140.4
Ope. Liquid	...	...	...	...	...	20758.8
Test Liquid	...	114132.5	...	...	...	...
Totals	27222.9	141355.4	27222.9	37426.7	37426.7	58185.5

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	27222.9 kgm
Shop Test Wt.	- Fabricated weight + water (full)	141355.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	27222.9 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	37426.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	37426.7 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	58185.5 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	56360.0 kgm
Field Test Wt.	- Empty weight + test medium (full)	151559.1 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	14202853	13983931
20	30	33589908	33175218
30	40	33589908	33175218
40	50	27389160	27203682
50	60	22267608	22116812
60	70	14265717	13983931

Total	145305152.0	143638784.0 mm ²
Total	145.3	143.6 m ²



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N11-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N10A-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N10B-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N3-24''	1.60	1.96	145	150	GR 1.1	1.469	1.599	1.599	0.930
M1-24	1.60	1.96	145	150	GR 1.1	...	...	...	...
N4-24''	1.60	1.96	145	150	GR 1.1	1.469	1.599	1.599	0.930
N5-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N6-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N9-3''	4.53	5.10	145	300	GR 1.1	4.878	1.912	4.526	1.912
N1-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N2-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N7-2''	4.53	5.10	145	300	GR 1.1	6.338	0.452	4.526	0.452
M2-24''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N8A-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N8B-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
N12A	1.60	1.96	145	150	GR 1.1	...	...	...	...
N12B-2''	1.60	1.96	145	150	GR 1.1	...	...	...	...
Min Rating	0.452	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3-24''	657.225	1.20	75603424	38398	0.452	1.599	Pass
N4-24''	657.225	1.20	75603424	38398	0.452	1.599	Pass
N9-3''	111.125	0.50	1181304	4800	0.452	1.912	Pass
N7-2''	77.851	0.50	525024	3200	0.452	0.452	Pass

Evaluating Flange on Nozzle [N3-24'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(75603424) + 4 \cdot \text{abs}(38398)657.225 \leq \pi \cdot 657.225^3 \cdot ((1.599 - 0.452) + 1.200 \cdot 1.599)$$

$$= 0.13106E+10 \leq 0.27354E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-24'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.599, 1.200 \cdot 1.599 + 1.599 - 1310640896.0 / (\pi \cdot 657.225^3))$$

$$= \min(1.599, 2.049)$$

$$= 1.599 \text{ MPa}$$

Evaluating Flange on Nozzle [N4-24'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(75603424) + 4 \cdot \text{abs}(38398)657.225 \leq \pi \cdot 657.225^3 \cdot ((1.599 - 0.452) + 1.200 \cdot 1.599)$$

$$= 0.13106E+10 \leq 0.27354E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-24'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.599, 1.200 \cdot 1.599 + 1.599 - 1310640896.0 / (\pi \cdot 657.225^3))$$

$$= \min(1.599, 2.049)$$



= 1.599 MPa

Evaluating Flange on Nozzle [N9-3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800) 111.125 \leq \pi \cdot 111.125^3 [(4.526 - 0.452) + .500 \cdot 4.526]$$

$$= 0.21035E+08 \leq 0.27330E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N9-3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.526, 0.500 \cdot 4.526 + 4.526 - 21035196.0 / (\pi \cdot 111.125^3))$$

$$= \min(4.526, 1.912)$$

$$= 1.912 \text{ MPa}$$

Evaluating Flange on Nozzle [N7-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(525024) + 4 \cdot \text{abs}(3200) 77.851 \leq \pi \cdot 77.851^3 [(4.526 - 0.452) + .500 \cdot 4.526]$$

$$= 0.93972E+07 \leq 0.93972E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N7-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.526, 0.500 \cdot 4.526 + 4.526 - 9397238.0 / (\pi \cdot 77.851^3))$$

$$= \min(4.526, 0.452)$$

$$= 0.452 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	0	
Base Elevation	0	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.972) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$$

$$= \min(0.85, 0.911)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.736
Structure Height to Diameter ratio	3.135

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:

Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Z_g = 589788$ mm

Effective Height $[z]$:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 2300.0 + 0.0 = 2300.0 \text{ mm}$$

$$= 7.546 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z $[K_z]$:

$$\text{Because } z (7.546 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.41(15 / Z_g)^{2/\alpha}$$

$$= 2.41(15 / 1935.0)^{2/11.5}$$

$$= 1.035$$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:

= 1, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:



$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf} [1.587] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.736 \cdot 33.816$$

$$= 700.9 \text{ lbs.} [3117.6] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
EH-1	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-1	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-2	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-3	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
SHELL-4	2300.0	0.000	0.000	0.000	1.035	1.000	1.587
EH-2	2300.0	0.000	0.000	0.000	1.035	1.000	1.587

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLATFORM	15841300.00	2300.00	1.59	25141.70	1.30

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
cleat weight	0.00	2300.00	1.59	0.00
LL-1	0.00	3900.00	1.63	0.00
M1 davit	0.00	4160.00	1.64	0.00
VB	0.00	0.00	1.59	0.00
LL-2	0.00	3900.00	1.63	0.00
M2-davit	0.00	2300.00	1.59	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	2300	4492.8	3141646	1.58714	3429.32
20	30	2300	4516.8	13550398	1.58714	42447
30	40	2300	4516.8	13550398	1.58714	14791.2
40	50	2300	4492.8	11052288	1.58714	12064.3
50	60	2300	4492.8	8985600	1.58714	9808.39
60	70	2300	4502.4	3160146	1.58714	3449.51

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * S_{ms} = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * S_{m1} = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:

= $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)

= $0.020(13.320^{0.75})$

= 0.139 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:

= (1/Natural Frequency) = (1/33.0)

= 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:

= Minimum Value of ($1.7 \cdot 0.139$, 0.03) per 12.8.2

= 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:

= $0.3 \cdot S_{ds} \cdot W \cdot I$

= $0.3 \cdot 0.217 \cdot 536129 \cdot 1.5$

= 52433.465 N

Vertical load per 12.4-4, [Ev]:

= $0.2 \cdot S_{ds} \cdot W$

= $0.2 \cdot 0.217 \cdot 536129$

= 23303.8 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	1760	67016.2	6554.18
20	30	1760	67016.2	6554.18
20	30	1760	67016.2	6554.18
30	40	1760	67016.2	6554.18
40	50	1760	67016.2	6554.18



50	0	1760	67016.2	6554.18
50	60	1760	67016.2	6554.18
60	70	1760	67016.2	6554.18

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.

Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5280.088 mm
Center of Gravity of the Platforms	5330.000 mm
Center of Gravity of the Liquid	5280.000 mm
Center of Gravity of the Insulation	5252.589 mm
Center of Gravity of the Stiffening Rings	6999.315 mm
Center of Gravity of the Nozzles	4728.342 mm
Center of Gravity of the Added Weights (Operating)	4587.798 mm
Center of Gravity of the Added Weights (Empty)	4587.798 mm
Center of Gravity of Bare Shell New and Cold	4836.505 mm
Center of Gravity of Bare Shell Corroded	4794.657 mm
Vessel CG in the Operating Condition	5115.395 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4929.485 mm
Vessel CG in the Test Condition	5212.438 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

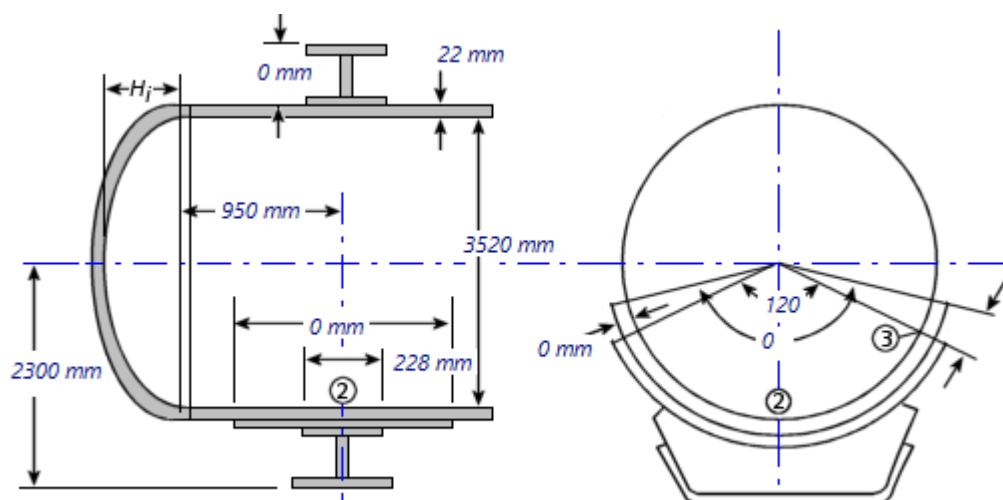
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1771.75 \cdot 20.5}, 2 \cdot 950.0)$$

$$= 525.3054 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1771.75	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	886.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	386538.91	N
Pressure used in Analysis	P	0.350 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	23.84	138.00
Long. Stress at Bottom of Midspan	35.22	138.00
Long. Stress at Top of Saddles	16.42	138.00
Long. Stress at Bottom of Saddles	14.41	138.00
Tangential Shear in Shell	9.19	110.40



Circ. Stress at Horn of Saddle	38.76	172.50
Circ. Compressive Stress in Shell	2.73	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (85989.8 / 2 + 0) \cdot 2300.0 / 3080.0$$

$$= 96319.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(36062.26, 0) \cdot 2300.0 / 8560.0$$

$$= 9689.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(52433.46, 0) \cdot 2300.0 / 8560.0$$

$$= 14088.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (52433 / 2 + 0) \cdot 2300.0 / 3080.0$$

$$= 58732.3 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 290219 + \max(9690, 96320, 14088, 58732)$$

$$= 386538.9 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi / 4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169 (4.633 + \pi / 4 (3.764 \cdot 1.2)^2)$$

$$= 36064.121 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 394806.81 N

Transverse Shear Load 42994.89 N

Longitudinal Shear Load 52433.46 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0161 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2) / (2a \cdot L)) / (1 + (4h_m) / (3L))]$$

$$= -386539 \cdot 950.0 [1 - (1 - 950.0 / 10560.0 + (1771.75^2 - 886.75^2) /$$

$$(2 \cdot 950.0 \cdot 10560.0)) / (1 + (4 \cdot 886.75) / (3 \cdot 10560.0))]$$

$$= -27969974.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2) / (L^2)) / (1 + (4h_m) / (3L)) - 4a/L)$$

$$= 386539 \cdot 10560 / 4 ((1 + 2(1772^2 - 887^2) / (10560^2)) / (1 + (4 \cdot 887) /$$

$$(3 \cdot 10560)) - 4 \cdot 950 / 10560)$$

$$= 589464192.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 0.35 \cdot 1771.75 / (2 \cdot 10.5) - 0.58946 \text{E}+09 / (\pi \cdot 1771.8^2 \cdot 10.5)$$



$$= 23.84 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.35 \cdot 1771.75 / (2 \cdot 10.5) + 0.58946E+09 / (\pi \cdot 1771.8^2 \cdot 10.5)$$

$$= 35.22 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.35 \cdot 1771.75 / (2 \cdot 20.5) - 27969974.0 / (0.1066 \cdot \pi \cdot 1771.8^2 \cdot 20.5)$$

$$= 16.42 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.35 \cdot 1771.75 / (2 \cdot 20.5) + 27969974.0 / (0.1923 \cdot \pi \cdot 1771.8^2 \cdot 20.5)$$

$$= 14.41 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 386539(10560.0 - 2 \cdot 950.0) / (10560.0 + (4 \cdot 886.75 / 3))$$

$$= 285073.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 285073.41 / (1771.75 \cdot 20.5)$$

$$= 9.19 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1771.75 \cdot 20.5)}$$

$$= 148.653 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 386539 \cdot 0.1 / (20.5(228.0 + 148.65 + 148.65))$$

$$= -2.73 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -386539 / (4 \cdot 20.5(228.0 + 148.653 + 148.653)) -$$

$$12 \cdot 0.016 \cdot 386539 \cdot 1771.75 / (10560.0 \cdot 20.5^2)$$

$$= -38.76 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1771.75 \cdot 20.5)}, 2 \cdot 950.0)$$

$$= 525.31 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 8560.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 8560.0(145.0 - 21.1)$$

$$= 13.143 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	498.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1761.5 \cdot 20.5)} + 300.0) 20.5^3) / 12$$

$$= 428202.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 428202.938 + 12227.104 (178.904 - 10.25)^2$$

$$= 348218176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	596.4	20.5	10.2	12227.1	125327.8	0.348E+09
Wearplate	300.0	10.0	25.5	3000.0	76500.0	0.706E+08
Web	16.0	488.0	274.5	7808.0	2143299.0	0.226E+09
BasePlate	254.0	20.0	528.5	5080.0	2684781.8	0.621E+09
Totals	...	...	...	28115.1	5029908.5	0.127E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5029909 / 28115.107$$

$$= 178.904 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 386538.906$$

$$= 78669.0469 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.9519 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1761.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 843.251 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 66364676.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.3739 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(386539 + 8268)254.0) / (4 \cdot 3080.0 \cdot 137.997)}$$

$$= 13.303 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 3103.8347 / (4 - 1)$$



$$= 1034.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 1034.6116 \cdot 254.0 / 2$$

$$= 131395.672 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 394806.844 / (3080.0 \cdot 254.0)$$

$$= 0.505 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 131395.7 \cdot 0.5$$

$$= 66310.344 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 3392.0 + 8276.893$$

$$= 11668.893 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 66310.3 / 11668.8926$$

$$= 5.683 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 11668.893$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(15803123 / 11668.893)}$$

$$= 36.801 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 2300.0 - \cos(120.0 / 2) (1760.0 + 22.0 + 10.0) - 20.0$$

$$= 1384.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120.0 / 2) (1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 3080.0 - 50.8$$

$$= 3029.200 \text{ mm}$$



Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (3103.835 - 3029.2) / 2$$

$$= 37.317 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1384.0^2 + 37.317^2}$$

$$= 1384.503 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713 / 207.0)}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1384.503 / 36.801$$

$$= 37.622$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 52433.5 / (2 \cdot 3080.0) \cdot 1034.612 \cdot 1384.5 / 2$$

$$= 6098810.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($37.6216 < 139.0351$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r)^2 / (8 \cdot Cc) - (KL/r)^2 / (8 \cdot Cc^3))$$

$$= (1 - (37.62)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (37.62)^2 / (8 \cdot 139.04) - (37.62^3) / (8 \cdot 139.04^3))$$

$$= 112.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.68 / 112.94 + (6098810 \cdot 114.0 / 15803138) / 138.0$$

$$= 0.369$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1034.6	16.0	0.0	16553.8	0.0	0.353E+06
Totals	...	...	...	19945.8	...	0.162E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 19945.785$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpdthk$$

$$= 2300.0 - \sqrt{(1792.0 + 10.0)^2 - (1034.612/2)^2} - 20.0$$

$$= 564.291 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(16150807 / 19945.785)}$$

$$= 28.456 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 564.291 / 28.456$$



$$= 19.830$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 52433.457 / (2 \cdot 3080.0)$$

$$= 8.512 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.512 \cdot 1034.612 \cdot 564.291$$

$$= 4971463.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4971463 / 141673.875$$

$$= 35.080 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($19.8304 < 139.0351$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.83)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (19.83) / (8 \cdot 139.04) - (19.83^3) / (8 \cdot 139.04^3))$$

$$= 119.1 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.65 / 119.14 + (4971463 \cdot 114.0 / 16150807) / 138.0$$

$$= 0.310$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	90.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BlArea	477.5988 mm ²
Saddle Load QO (Weight)	QO	298487.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	14088.4 N
Maximum Transverse Force	Ft	42994.9 N
Maximum Longitudinal Force	Fl	52433.5 N
Saddle Bolted to Steel Foundation	No	

Friction force due to thermal growth, [Ff]:

$$= 166489.4 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 166489.4 / (477.599 \cdot 4)$$

$$= 87.2 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 42994.9 / (477.599 \cdot 4)$$

$$= 22.5 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 52433.46 / (100.0 \cdot 4.0)$$

$$= 131.0948 \text{ mm}^2$$

**Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 42994.89 + 0.0$$

$$= 98928344.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 98928344 / 298487.06$$

$$= 331.30 \text{ mm} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 131.0948, 0.0]$$

$$= 131.0948 \text{ mm}^2$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 10.0, c = 1.5 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.287$, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -46 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1766.7499 \cdot 10.5}, 2 \cdot 950.0)$$

$$= 440.4745 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1766.75	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	888.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		365533.84	N
Pressure used in Analysis	P	0.350	MPa

Horizontal Vessel Analysis Results:	Actual	Allowable
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	MPa	MPa
Long. Stress at Top of Midspan	24.04	138.00
Long. Stress at Bottom of Midspan	34.85	138.00
Long. Stress at Top of Saddles	31.89	138.00
Long. Stress at Bottom of Saddles	28.09	138.00
Tangential Shear in Shell	17.01	110.40
Circ. Stress at Horn of Saddle	127.62	172.50
Circ. Compressive Stress in Shell	6.01	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(85989.8/2 + 0) \cdot 2300.0/3080.0$$

$$= 96319.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(35765.77, 0) \cdot 2300.0/8560.0$$

$$= 9610.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(52433.46, 0) \cdot 2300.0/8560.0$$

$$= 14088.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(52433/2 + 0) \cdot 2300.0/3080.0$$

$$= 58732.3 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 269214 + \max(9610, 96320, 14088, 58732)$$

$$= 365533.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(4.633 + \pi/4(3.744 \cdot 1.2)^2)$$

$$= 35767.621 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	373802.09	N
Transverse Shear Load	42994.89	N
Longitudinal Shear Load	52433.46	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0162	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -365534 \cdot 950.0 [1 - (1 - 950.0/10560.0 + (1766.75^2 - 888.75^2)/(2 \cdot 950.0 \cdot 10560.0))/(1 + (4 \cdot 888.75)/(3 \cdot 10560.0))]$$

$$= -26853578.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:



$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\
 &= 365534 \cdot 10560 / 4 \cdot ((1 + 2(1767^2 - 889^2) / (10560^2)) / (1 + (4 \cdot 889) / (3 \cdot 10560)) - 4 \cdot 950 / 10560) \\
 &= 556895808.0 \text{ N-mm}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 0.35 \cdot 1766.75 / (2 \cdot 10.5) - 0.55690E+09 / (\pi \cdot 1766.7^2 \cdot 10.5) \\
 &= 24.04 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 0.35 \cdot 1766.75 / (2 \cdot 10.5) + 0.55690E+09 / (\pi \cdot 1766.7^2 \cdot 10.5) \\
 &= 34.85 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.35 \cdot 1766.75 / (2 \cdot 10.5) - 26853578.0 / (0.1066 \cdot \pi \cdot 1766.7^2 \cdot 10.5) \\
 &= 31.89 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0.35 \cdot 1766.75 / (2 \cdot 10.5) + 26853578.0 / (0.1923 \cdot \pi \cdot 1766.7^2 \cdot 10.5) \\
 &= 28.09 \text{ MPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 365534(10560.0 - 2 \cdot 950.0) / (10560.0 + (4 \cdot 888.75 / 3)) \\
 &= 269520.9 \text{ N}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 269520.91 / (1766.7499 \cdot 10.5) \\
 &= 17.01 \text{ MPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1766.75 \cdot 10.5)} \\
 &= 106.237 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 365534 \cdot 0.1 / (10.5(228.0 + 106.24 + 106.24)) \\
 &= -6.01 \text{ MPa}
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ_{*7}]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2) \\
 &= -365534 / (4 \cdot 10.5(228.0 + 106.237 + 106.237)) - 12 \cdot 0.016 \cdot 365534 \cdot 1766.75 / (10560.0 \cdot 10.5^2) \\
 &= -127.62 \text{ MPa}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(228.0 + 1.56 \cdot \sqrt{(1766.75 \cdot 10.5)}, 2 \cdot 950.0) \\
 &= 440.47 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	508.0	mm
Friction Coefficient	mu	0.600	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1761.5 \cdot 10.5)} + 300.0)10.5^3)/12$$

$$= 49407.250 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 49407.25 + 5377.665(229.019 - 5.25)^2$$

$$= 269323552.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	512.2	10.5	5.2	5377.7	28232.7	0.269E+09
Wearplate	300.0	10.0	15.5	3000.0	46500.0	0.137E+09
Web	16.0	498.0	269.5	7968.0	2147378.8	0.178E+09
BasePlate	254.0	20.0	528.5	5080.0	2684781.8	0.456E+09
Totals	...	...	...	21425.7	4906893.0	0.104E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4906893/21425.668$$

$$= 229.019 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 365533.844$$

$$= 74394.0703 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.6361 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1761.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 843.251 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 62758340.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13.8204 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(365534 + 8268)254.0 / (4 \cdot 3080.0 \cdot 137.997))}$$

$$= 12.944 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1760.0 + 12.0 + 10.0)$$

$$= 3086.514 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$



$$= 3086.5142 / (4 - 1)$$

$$= 1028.838 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 1028.838 \cdot 254.0 / 2$$

$$= 130662.430 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 373802.094 / (3080.0 \cdot 254.0)$$

$$= 0.478 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 130662.4 \cdot 0.48$$

$$= 62432.117 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 3392.0 + 8230.704$$

$$= 11622.704 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 62432.1 / 11622.7041$$

$$= 5.372 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 11622.704$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15803123 / 11622.704}$$

$$= 36.874 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300.0 - \cos(120.0 / 2) (1760.0 + 12.0 + 10.0) - 20.0$$

$$= 1389.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (1760.0 + 12.0 + 10.0)$$

$$= 3086.514 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - \text{clearance}$$

$$= 3080.0 - 50.8$$



$$= 3029.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (L_w - B_d) / 2 \\ &= (3086.514 - 3029.2) / 2 \\ &= 28.657 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{L_v^2 + D_d^2} \\ &= \sqrt{1389.0^2 + 28.657^2} \\ &= 1389.296 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 202713 / 207.0)} \\ &= 139.035 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 1389.296 / 36.874 \\ &= 37.677 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\ &= 52433.5 / (2 \cdot 3080.0) \cdot 1028.838 \cdot 1389.3 / 2 \\ &= 6085770.000 \text{ N-mm} \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($37.677 < 139.0351$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (37.68)^2 / (2 \cdot 139.04^2)) 207 / \\ &\quad (5/3 + 3 \cdot (37.68) / (8 \cdot 139.04) - (37.68^3) / (8 \cdot 139.04^3)) \\ &= 112.9 \text{ MPa} \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\ &= 5.37 / 112.92 + (6085770 \cdot 114.0 / 15803138) / 138.0 \\ &= 0.366 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1028.8	16.0	0.0	16461.4	0.0	0.351E+06
Totals	...	...	...	19853.4	...	0.161E+08

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0 / 19853.408 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 228.0 / 2 \\ &= 114.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk} \\ &= 2300.0 - \sqrt{(1782.0 + 10.0)^2 - (1028.838/2)^2} - 20.0 \\ &= 573.865 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\ &= \sqrt{16148837 / 19853.408} \\ &= 28.520 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



$$= 1 \cdot 573.865 / 28.52$$

$$= 20.121$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 52433.457 / (2 \cdot 3080.0)$$

$$= 8.512 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.512 \cdot 1028.838 \cdot 573.865$$

$$= 5027598.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5027598 / 141656.578$$

$$= 35.480 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($20.1213 < 139.0351$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20.12)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (20.12) / (8 \cdot 139.04) - (20.12^3) / (8 \cdot 139.04^3))$$

$$= 119 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 6.29 / 119.05 + (5027598 \cdot 114.0 / 16148837) / 138.0$$

$$= 0.310$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	90.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	477.5988 mm ²
Saddle Load Q0 (Weight)	Q0	277482.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	14088.4 N
Maximum Transverse Force	Ft	42994.9 N
Maximum Longitudinal Force	Fl	52433.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 52433.5 / (477.599 \cdot 4)$$

$$= 27.4 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 42994.9 / (477.599 \cdot 4)$$

$$= 22.5 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 52433.46 / (100.0 \cdot 4.0)$$

$$= 131.0948 \text{ mm}^2$$

Bolt Area due to Transverse Load:



Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 42994.89 + 0.0$$

$$= 98928344.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 98928344 / 277482.38$$

$$= 356.38 \text{ mm} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 131.0948, 0.0]$$

$$= 131.0948 \text{ mm}^2$$

MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 1.5$ mm, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.561$, Temp. Reduction = 26 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

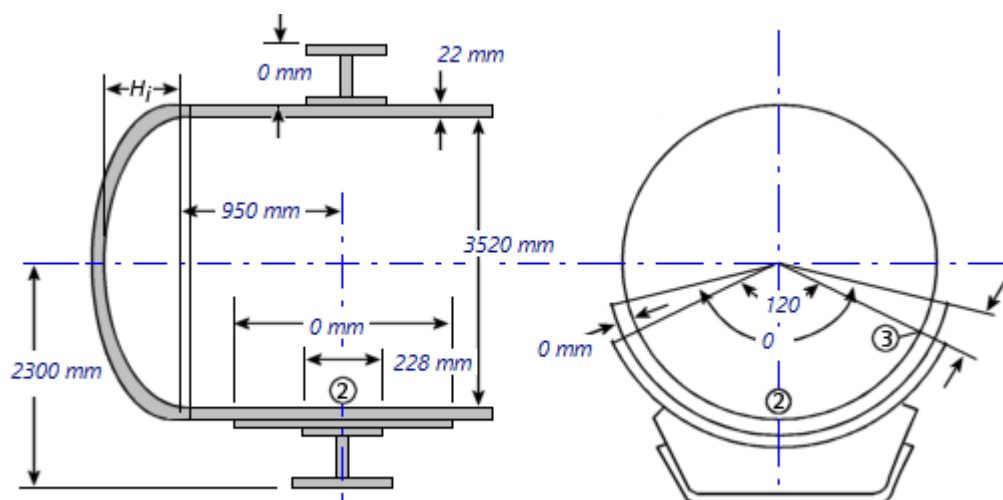
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1771.0 \cdot 22.0}, 2 \cdot 950.0)$$

$$= 535.9252 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1771.00	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	886.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	777111.12	N
Pressure used in Analysis	P	0.605 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	34.61	234.00
Long. Stress at Bottom of Midspan	54.66	234.00
Long. Stress at Top of Saddles	26.78	234.00
Long. Stress at Bottom of Saddles	23.00	234.00
Tangential Shear in Shell	17.22	138.58



Circ. Stress at Horn of Saddle	68.51	350.99
Circ. Compressive Stress in Shell	5.01	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (28376.6/2 + 0) \cdot 2300.0/3080.0$$

$$= 31785.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(11900.55, 0) \cdot 2300.0/8560.0$$

$$= 3197.6 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 745326 + \max(3198, 31786, 0, 0)$$

$$= 777111.1 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169 (4.633 + \pi/4 (3.764 \cdot 1.2)^2)$$

$$= 36064.121 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	785379.06	N
Transverse Shear Load	14188.31	N
Longitudinal Shear Load	11900.55	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0161	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -777111 \cdot 950.0 [1 - (1 - 950.0/10560.0 + (1771.0^2 - 886.0^2) / (2 \cdot 950.0 \cdot 10560.0)) / (1 + (4 \cdot 886.0) / (3 \cdot 10560.0))]$$

$$= -56217588.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 777111 \cdot 10560 / 4 ((1 + 2(1771^2 - 886^2)/(10560^2)) / (1 + (4 \cdot 886) / (3 \cdot 10560)) - 4 \cdot 950 / 10560)$$

$$= 1185199104.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1771.0 / (2 \cdot 12.0) - 0.11852E+10 / (\pi \cdot 1771.0^2 \cdot 12.0)$$

$$= 34.61 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1771.0 / (2 \cdot 12.0) + 0.11852E+10 / (\pi \cdot 1771.0^2 \cdot 12.0)$$

$$= 54.66 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1771.0 / (2 \cdot 22.0) - 56217588.0 / (0.1066 \cdot \pi \cdot 1771.0^2 \cdot 22.0)$$



= 26.78 MPa

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1771.0 / (2 \cdot 22.0) + 56217588.0 / (0.1923 \cdot \pi \cdot 1771.0^2 \cdot 22.0)$$

$$= 23.00 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 777111(10560.0 - 2 \cdot 950.0) / (10560.0 + (4 \cdot 886.0 / 3))$$

$$= 573170.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 573170 / (1771.0 \cdot 22.0)$$

$$= 17.22 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1771.0 \cdot 22.0)}$$

$$= 153.963 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 777111 \cdot 0.1 / (22.0(228.0 + 153.96 + 153.96))$$

$$= -5.01 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -777111 / (4 \cdot 22.0(228.0 + 153.963 + 153.963)) -$$

$$12 \cdot 0.016 \cdot 777111 \cdot 1771.0 / (10560.0 \cdot 22.0^2)$$

$$= -68.51 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1771.0 \cdot 22.0)}, 2 \cdot 950.0)$$

$$= 535.93 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	205.0	MPa
Height of Web at Center	Hw,c	498.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1760.0 \cdot 22.0)} + 300.0)^3) / 12$$

$$= 538581.875 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 538582 + 13353.282(173.566 - 11.0)^2$$

$$= 353435392.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	607.0	22.0	11.0	13353.3	146886.1	0.353E+09
Wearplate	300.0	10.0	27.0	3000.0	81000.0	0.645E+08
Web	16.0	488.0	276.0	7808.0	2155009.5	0.237E+09
BasePlate	254.0	20.0	530.0	5080.0	2692401.0	0.646E+09



Totals	...	...	...	29241.3	5075296.5	0.130E+10
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 5075297/29241.285$$

$$= 173.566 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 777111$$

$$= 158158.9531 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.9555 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 122.9978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1760.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 844.492 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 133618104.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 17.8292 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 136.6642 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(777111 + 8268) \cdot 254.0) / (4 \cdot 3080.0 \cdot 137.997)}$$

$$= 18.763 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 3103.8347 / (4 - 1)$$

$$= 1034.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1034.6116 \cdot 254.0/2$$

$$= 131395.672 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 785379 / (3080.0 \cdot 254.0)$$

$$= 1.004 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 131395.7 \cdot 1.0$$



$$= 131909.453 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 3392.0 + 8276.893$$

$$= 11668.893 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 131909.5/11668.8926$$

$$= 11.305 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/11668.893$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0/2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15803123/11668.893)}$$

$$= 36.801 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300.0 - \cos(120.0/2)(1760.0 + 22.0 + 10.0) - 20.0$$

$$= 1384.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2)(1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3080.0 - 50.8$$

$$= 3029.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd)/2$$

$$= (3103.835 - 3029.2)/2$$

$$= 37.317 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1384.0^2 + 37.317^2}$$

$$= 1384.503 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713/205.0)}$$

$$= 139.712$$



Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 1384.503/36.801$$

$$= 37.622$$

Bending Moment $[Rm]$:

$$= Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 11900.5 / (2 \cdot 3080.0) \cdot 1034.612 \cdot 1384.5 / 2$$

$$= 1384214.750 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (37.6216 < 139.7117) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.62)^2 / (2 \cdot 139.71^2)) 205 /$$

$$(5/3 + 3 \cdot (37.62) / (8 \cdot 139.71) - (37.62^3) / (8 \cdot 139.71^3))$$

$$= 111.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 11.31 / 111.92 + (1384215 \cdot 114.0 / 15803138) / 136.66$$

$$= 0.174$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1034.6	16.0	0.0	16553.8	0.0	0.353E+06
Totals	...	...	...	19945.8	...	0.162E+08

Distance to Centroid from Datum $[y_{tot}]$:

$$= AY / A$$

$$= 0.0 / 19945.785$$

$$= 0.000 \text{ mm}$$

Distance to Centroid $[C1]$:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib $[L]$:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 2300.0 - \sqrt{ (1792.0 + 10.0)^2 - (1034.612/2)^2 } - 20.0$$

$$= 564.291 \text{ mm}$$

Radius of Gyration $[r]$:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (16150807 / 19945.785) }$$

$$= 28.456 \text{ mm}$$

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 564.291 / 28.456$$

$$= 19.830$$

Unit Force $[Force, u]$:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 11900.546 / (2 \cdot 3080.0)$$

$$= 1.932 \text{ N/mm}$$

Moment at base of inner Rib $[M_{base, c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.932 \cdot 1034.612 \cdot 564.291$$

$$= 1128346.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_{B, base, c}]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1128347 / 141673.875$$



= 7.962 MPa

Compressive Allowable, $KL/r < C_c$ (19.8304 < 139.7117) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.83)^2 / (2 \cdot 139.71^2)) 205 /$$

$$(5/3 + 3 \cdot (19.83) / (8 \cdot 139.71) - (19.83^3) / (8 \cdot 139.71^3))$$

$$= 118 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 13.22 / 118.02 + (1128347 \cdot 114.0 / 16150807) / 136.66$$

$$= 0.170$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	90.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	477.5988 mm ²
Saddle Load Q0 (Weight)	Q0	753593.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	3197.6 N
Maximum Transverse Force	Ft	14188.3 N
Maximum Longitudinal Force	F1	11900.5 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11900.5 / (477.599 \cdot 4)$$

$$= 6.2 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 14188.3 / (477.599 \cdot 4)$$

$$= 7.4 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 11900.55 / (68.4 \cdot 4.0)$$

$$= 43.5006 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 14188.31 + 0.0$$

$$= 32646354.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 32646354 / 753594$$

$$= 43.30 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 43.5006, 0.0]$$



= 43.5006 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{ 1766.0 \cdot 12.0 }, 2 \cdot 950.0)$$

$$= 455.0965 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1766.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	888.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		756111.94	N
Pressure used in Analysis	P	0.605	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	34.71	234.00
Long. Stress at Bottom of Midspan	54.30	234.00
Long. Stress at Top of Saddles	48.93	234.00
Long. Stress at Bottom of Saddles	42.05	234.00
Tangential Shear in Shell	30.80	138.58
Circ. Stress at Horn of Saddle	205.55	350.99
Circ. Compressive Stress in Shell	10.53	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (28376.6 / 2 + 0) \cdot 2300.0 / 3080.0$$

$$= 31785.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(11802.71, 0) \cdot 2300.0 / 8560.0$$

$$= 3171.3 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 724327 + \max(3171, 31786, 0, 0)$$

$$= 756111.9 \text{ N}$$



Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(4.633 + \pi/4(3.744 \cdot 1.2)^2)$$

$$= 35767.621 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	764380.19	N
Transverse Shear Load	14188.31	N
Longitudinal Shear Load	11802.71	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0162	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -756112 \cdot 950.0 [1 - (1 - 950.0/10560.0 + (1766.0^2 - 888.0^2)/(2 \cdot 950.0 \cdot 10560.0))/(1 + (4 \cdot 888.0)/(3 \cdot 10560.0))]$$

$$= -55533048.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 756112 \cdot 10560/4 ((1 + 2(1766^2 - 888^2)/(10560^2))/(1 + (4 \cdot 888)/(3 \cdot 10560)) - 4 \cdot 950/10560)$$

$$= 1152063616.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.605 \cdot 1766.0/(2 \cdot 12.0) - 0.11521\text{E}+10/(\pi \cdot 1766.0^2 \cdot 12.0)$$

$$= 34.71 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1766.0/(2 \cdot 12.0) + 0.11521\text{E}+10/(\pi \cdot 1766.0^2 \cdot 12.0)$$

$$= 54.30 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.605 \cdot 1766.0/(2 \cdot 12.0) - 55533048.0/(0.1066 \cdot \pi \cdot 1766.0^2 \cdot 12.0)$$

$$= 48.93 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.605 \cdot 1766.0/(2 \cdot 12.0) + 55533048.0/(0.1923 \cdot \pi \cdot 1766.0^2 \cdot 12.0)$$

$$= 42.05 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 756112(10560.0 - 2 \cdot 950.0)/(10560.0 + (4 \cdot 888.0/3))$$

$$= 557555.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 557555/(1766.0 \cdot 12.0)$$

$$= 30.80 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1766.0 \cdot 12.0)}$$

$$= 113.548 \text{ mm}$$



Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 756112 \cdot 0.1 / (12.0(228.0 + 113.55 + 113.55))$$

$$= -10.53 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -756112 / (4 \cdot 12.0(228.0 + 113.548 + 113.548)) -$$

$$12 \cdot 0.016 \cdot 756112 \cdot 1766.0 / (10560.0 \cdot 12.0^2)$$

$$= -205.55 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1766.0 \cdot 12.0}, 2 \cdot 950.0)$$

$$= 455.10 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	205.0	MPa
Height of Web at Center	Hw,c	508.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{1760.0 \cdot 12.0} + 300.0)^3) / 12$$

$$= 75846.227 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 75846.227 + 6320.525(220.875 - 6.0)^2$$

$$= 291902848.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	526.7	12.0	6.0	6320.5	37923.1	0.292E+09
Wearplate	300.0	10.0	17.0	3000.0	51000.0	0.125E+09
Web	16.0	498.0	271.0	7968.0	2159329.0	0.185E+09
BasePlate	254.0	20.0	530.0	5080.0	2692401.0	0.486E+09
Totals	...	...	...	22368.5	4940653.5	0.109E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4940654 / 22368.527$$

$$= 220.875 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 756112$$



$$= 153885.1719 \text{ N}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 9.5899 \text{ MPa} \\ \text{Allowed Stress, } Sa &= 0.6 * \text{Yield Str} = 122.9978 \text{ MPa} \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R * \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 2300.0 - 1760.0 * \sin(120.0/2) / 1.0472 \\ &= 844.492 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh * d = 130007464.0000 \text{ N-mm}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M * C1 / I) = 26.4105 \text{ MPa} \\ \text{Allowed Stress, } Sa &= 2/3 * \text{Yield Str} = 136.6642 \text{ MPa} \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress})} \\ &= \sqrt{3(756112 + 8268)254.0 / (4 * 3080.0 * 137.997)} \\ &= 18.510 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 * \cos(90 - 120.0/2) (1760.0 + 12.0 + 10.0) \\ &= 3086.514 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 3086.5142 / (4 - 1) \\ &= 1028.838 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e * \text{Bpwid} / 2 \\ &= 1028.838 * 254.0 / 2 \\ &= 130662.430 \text{ mm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth}) \\ &= 764380 / (3080.0 * 254.0) \\ &= 0.977 \text{ N/mm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap * Bp \\ &= 130662.4 * 0.98 \\ &= 127666.141 \text{ N} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 3392.0 + 8230.704 \\ &= 11622.704 \text{ mm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 127666.1 / 11622.7041 \\ &= 10.985 \text{ MPa} \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08



Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 228.0 - 16.0 \\
 &= 212.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/11622.704 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 228.0/2 \\
 &= 114.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(15803123/11622.704)} \\
 &= 36.874 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2300.0 - \cos(120.0/2) (1760.0 + 12.0 + 10.0) - 20.0 \\
 &= 1389.000 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120.0/2) (1760.0 + 12.0 + 10.0) \\
 &= 3086.514 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 3080.0 - 50.8 \\
 &= 3029.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (3086.514 - 3029.2) / 2 \\
 &= 28.657 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{1389.0^2 + 28.657^2} \\
 &= 1389.296 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 202713/205.0)} \\
 &= 139.712
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1389.296/36.874 \\
 &= 37.677
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 11802.7 / (2 \cdot 3080.0) \cdot 1028.838 \cdot 1389.3/2 \\
 &= 1369899.250 \text{ N-mm}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($37.677 < 139.7117$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (37.68)^2 / (2 \cdot 139.71^2)) 205 / \\
 &\quad (5/3 + 3 \cdot (37.68) / (8 \cdot 139.71) - (37.68^3) / (8 \cdot 139.71^3)) \\
 &= 111.9 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 10.99/111.9 + (1369899 \cdot 114.0/15803138) / 136.66
 \end{aligned}$$



= 0.170

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1028.8	16.0	0.0	16461.4	0.0	0.351E+06
Totals	...	...	...	19853.4	...	0.161E+08

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/19853.408

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 228.0/2

= 114.000 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 2300.0 - $\sqrt{(1782.0 + 10.0)^2 - (1028.838/2)^2}$ - 20.0

= 573.865 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{16148837/19853.408}$

= 28.520 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 573.865/28.52

= 20.121

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 11802.706/(2 • 3080.0)

= 1.916 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.916 • 1028.838 • 573.865

= 1131706.000 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1131706/141656.578

= 7.987 MPa

Compressive Allowable, KL/r < Cc (20.1213 < 139.7117) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (20.12)^2 / (2 \cdot 139.71^2)) 205 /$

$(5/3 + 3 \cdot (20.12) / (8 \cdot 139.71) - (20.12^3) / (8 \cdot 139.71^3))$

= 117.9 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 12.86/117.93 + (1131706 • 114.0/16148837)/136.66

= 0.167

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-36	
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm



Distance from Bolts to Edge	Edgedis	90.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	477.5988 mm ²
Saddle Load Q0 (Weight)	Q0	732594.7 N
Saddle Load QL (Wind/Seismic contribution)	QL	3171.3 N
Maximum Transverse Force	Ft	14188.3 N
Maximum Longitudinal Force	Fl	11802.7 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11802.7 / (477.599 \cdot 4)$$

$$= 6.2 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 14188.3 / (477.599 \cdot 4)$$

$$= 7.4 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 11802.71 / (68.4 \cdot 4.0)$$

$$= 43.1430 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2300.0 \cdot 14188.31 + 0.0$$

$$= 32646354.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 32646354 / 732595$$

$$= 44.54 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 43.143, 0.0]$$

$$= 43.1430 \text{ mm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

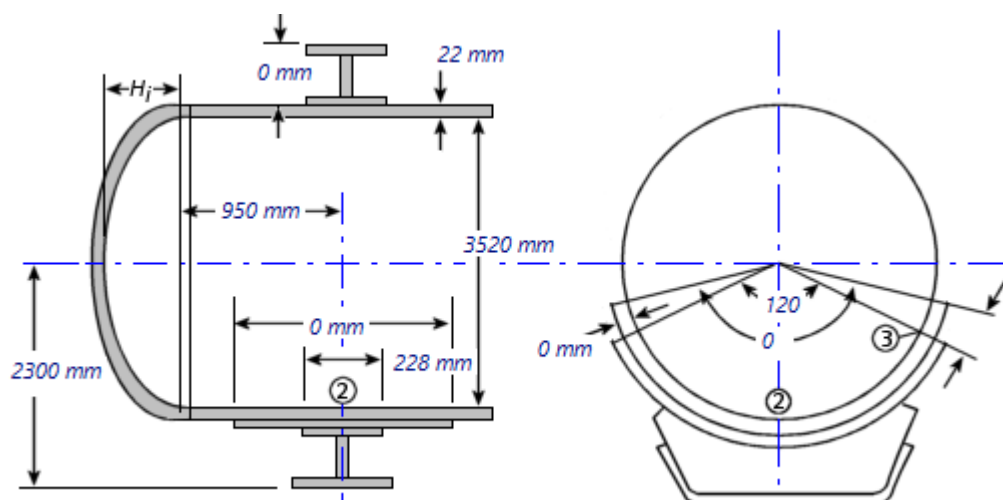
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1771.75 \cdot 20.5}, 2 \cdot 950.0)$$

$$= 525.3054 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1771.75	mm
Shell Thickness used in this Case	t	22.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	886.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		386538.91	N
Pressure used in Analysis	P	-0.101	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top	of Midspan	-14.24	-87.08
Long. Stress at Top	of Midspan	14.24	138.00
Long. Stress at Bottom	of Midspan	-2.86	-87.08
Long. Stress at Bottom	of Midspan	2.86	138.00
Long. Stress at Top	of Saddles	-3.08	-169.54
Long. Stress at Top	of Saddles	3.08	138.00



Long. Stress at Bottom of Saddles	-5.10	-169.54
Long. Stress at Bottom of Saddles	5.10	138.00
Tangential Shear in Shell	9.19	110.40
Circ. Stress at Horn of Saddle	38.76	172.50
Circ. Compressive Stress in Shell	2.73	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(85989.8/2 + 0) \cdot 2300.0/3080.0$$

$$= 96319.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(36062.26, 0) \cdot 2300.0/8560.0$$

$$= 9689.6 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(52433.46, 0) \cdot 2300.0/8560.0$$

$$= 14088.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(52433/2 + 0) \cdot 2300.0/3080.0$$

$$= 58732.3 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 290219 + \max(9690, 96320, 14088, 58732)$$

$$= 386538.9 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(4.633 + \pi/4(3.764 \cdot 1.2)^2)$$

$$= 36064.121 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 394806.81 N

Transverse Shear Load 42994.89 N

Longitudinal Shear Load 52433.46 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0161 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -386539 \cdot 950.0 [1 - (1 - 950.0/10560.0 + (1771.75^2 - 886.75^2) /$$

$$(2 \cdot 950.0 \cdot 10560.0) / (1 + (4 \cdot 886.75) / (3 \cdot 10560.0))]$$

$$= -27969974.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 386539 \cdot 10560/4 [(1 + 2(1772^2 - 887^2)/(10560^2)) / (1 + (4 \cdot 887) /$$

$$(3 \cdot 10560)) - 4 \cdot 950/10560]$$

$$= 589464192.0 \text{ N-mm}$$



Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1771.75 / (2 \cdot 10.5) - 0.58946E+09 / (\pi \cdot 1771.8^2 \cdot 10.5)$$

$$= -14.24 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1771.75 / (2 \cdot 10.5) + 0.58946E+09 / (\pi \cdot 1771.8^2 \cdot 10.5)$$

$$= -2.86 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1771.75 / (2 \cdot 20.5) - 27969974.0 / (0.1066 \cdot \pi \cdot 1771.8^2 \cdot 20.5)$$

$$= -3.08 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1771.75 / (2 \cdot 20.5) + 27969974.0 / (0.1923 \cdot \pi \cdot 1771.8^2 \cdot 20.5)$$

$$= -5.10 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 386539(10560.0 - 2 \cdot 950.0) / (10560.0 + (4 \cdot 886.75 / 3))$$

$$= 285073.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 285073.41 / (1771.75 \cdot 20.5)$$

$$= 9.19 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1771.75 \cdot 20.5)}$$

$$= 148.653 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 386539 \cdot 0.1 / (20.5(228.0 + 148.65 + 148.65))$$

$$= -2.73 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -386539 / (4 \cdot 20.5(228.0 + 148.653 + 148.653)) -$$

$$12 \cdot 0.016 \cdot 386539 \cdot 1771.75 / (10560.0 \cdot 20.5^2)$$

$$= -38.76 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1771.75 \cdot 20.5)}, 2 \cdot 950.0)$$

$$= 525.31 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	498.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$



$$= ((1.56 \cdot \sqrt{(1761.5 \cdot 20.5)} + 300.0) 20.5^3) / 12$$

$$= 428202.938 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 428202.938 + 12227.104 (178.904 - 10.25)^2$$

$$= 348218176.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	596.4	20.5	10.2	12227.1	125327.8	0.348E+09
Wearplate	300.0	10.0	25.5	3000.0	76500.0	0.706E+08
Web	16.0	488.0	274.5	7808.0	2143299.0	0.226E+09
BasePlate	254.0	20.0	528.5	5080.0	2684781.8	0.621E+09
Totals	...	...	...	28115.1	5029908.5	0.127E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5029909 / 28115.107$$

$$= 178.904 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 386538.906$$

$$= 78669.0469 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.9519 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1761.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 843.251 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 66364676.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.3739 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(386539 + 8268)254.0 / (4 \cdot 3080.0 \cdot 137.997))}$$

$$= 13.303 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 3103.8347 / (4 - 1)$$

$$= 1034.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



$$= 1034.6116 \cdot 254.0/2$$

$$= 131395.672 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 394806.844 / (3080.0 \cdot 254.0)$$

$$= 0.505 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 131395.7 \cdot 0.5$$

$$= 66310.344 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 3392.0 + 8276.893$$

$$= 11668.893 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 66310.3 / 11668.8926$$

$$= 5.683 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.0 - 16.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 11668.893$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15803123 / 11668.893)}$$

$$= 36.801 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2300.0 - \cos(120.0/2) (1760.0 + 22.0 + 10.0) - 20.0$$

$$= 1384.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (1760.0 + 22.0 + 10.0)$$

$$= 3103.835 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3080.0 - 50.8$$

$$= 3029.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (3103.835 - 3029.2) / 2$$



$$= 37.317 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1384.0^2 + 37.317^2}$$

$$= 1384.503 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 202713 / 207.0}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1384.503 / 36.801$$

$$= 37.622$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 52433.5 / (2 \cdot 3080.0) \cdot 1034.612 \cdot 1384.5 / 2$$

$$= 6098810.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($37.6216 < 139.0351$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.62)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (37.62) / (8 \cdot 139.04) - (37.62^3) / (8 \cdot 139.04^3))$$

$$= 112.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.68 / 112.94 + (6098810 \cdot 114.0 / 15803138) / 138.0$$

$$= 0.369$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1034.6	16.0	0.0	16553.8	0.0	0.353E+06
Totals	...	...	...	19945.8	...	0.162E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 19945.785$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 2300.0 - \sqrt{(1792.0 + 10.0)^2 - (1034.612/2)^2} - 20.0$$

$$= 564.291 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{16150807 / 19945.785}$$

$$= 28.456 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 564.291 / 28.456$$

$$= 19.830$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$



$$= 52433.457 / (2 \cdot 3080.0)$$

$$= 8.512 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.512 \cdot 1034.612 \cdot 564.291$$

$$= 4971463.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4971463 / 141673.875$$

$$= 35.080 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (19.8304 < 139.0351) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.83)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (19.83) / (8 \cdot 139.04) - (19.83)^3 / (8 \cdot 139.04^3))$$

$$= 119.1 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.65 / 119.14 + (4971463 \cdot 114.0 / 16150807) / 138.0$$

$$= 0.310$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (300.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{1766.7499 \cdot 10.5}, 2 \cdot 950.0)$$

$$= 440.4745 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1766.75	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	10560.00	mm
Distance from Saddle to Vessel tangent	a1 or a	950.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	888.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2300.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		365533.84	N
Pressure used in Analysis	P	-0.101	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top	of Midspan	-13.93	-87.08
Long. Stress at Top	of Midspan	13.93	138.00
Long. Stress at Bottom	of Midspan	-3.12	-87.08
Long. Stress at Bottom	of Midspan	3.12	138.00
Long. Stress at Top	of Saddles	-6.08	-87.08
Long. Stress at Top	of Saddles	6.08	138.00



Long. Stress at Bottom of Saddles	-9.88	-87.08
Long. Stress at Bottom of Saddles	9.88	138.00
Tangential Shear in Shell	17.01	110.40
Circ. Stress at Horn of Saddle	127.62	172.50
Circ. Compressive Stress in Shell	6.01	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(85989.8/2 + 0) \cdot 2300.0/3080.0$$

$$= 96319.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(35765.77, 0) \cdot 2300.0/8560.0$$

$$= 9610.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(52433.46, 0) \cdot 2300.0/8560.0$$

$$= 14088.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(52433/2 + 0) \cdot 2300.0/3080.0$$

$$= 58732.3 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 269214 + \max(9610, 96320, 14088, 58732)$$

$$= 365533.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(4.633 + \pi/4(3.744 \cdot 1.2)^2)$$

$$= 35767.621 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 373802.09 N

Transverse Shear Load 42994.89 N

Longitudinal Shear Load 52433.46 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 3080.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0162 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -365534 \cdot 950.0 [1 - (1 - 950.0/10560.0 + (1766.75^2 - 888.75^2) /$$

$$(2 \cdot 950.0 \cdot 10560.0) / (1 + (4 \cdot 888.75) / (3 \cdot 10560.0))]$$

$$= -26853578.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm)/3L) - 4a/L)$$

$$= 365534 \cdot 10560/4 \cdot ((1 + 2(1767^2 - 889^2)/(10560^2)) / (1 + (4 \cdot 889) /$$

$$(3 \cdot 10560)) - 4 \cdot 950/10560)$$

$$= 556895808.0 \text{ N-mm}$$



Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -0.101 \cdot 1766.75 / (2 \cdot 10.5) - 0.55690E+09 / (\pi \cdot 1766.7^2 \cdot 10.5)$$

$$= -13.93 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1766.75 / (2 \cdot 10.5) + 0.55690E+09 / (\pi \cdot 1766.7^2 \cdot 10.5)$$

$$= -3.12 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.101 \cdot 1766.75 / (2 \cdot 10.5) - 26853578.0 / (0.1066 \cdot \pi \cdot 1766.7^2 \cdot 10.5)$$

$$= -6.08 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.101 \cdot 1766.75 / (2 \cdot 10.5) + 26853578.0 / (0.1923 \cdot \pi \cdot 1766.7^2 \cdot 10.5)$$

$$= -9.88 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 365534(10560.0 - 2 \cdot 950.0) / (10560.0 + (4 \cdot 888.75 / 3))$$

$$= 269520.9 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 269520.91 / (1766.7499 \cdot 10.5)$$

$$= 17.01 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1766.75 \cdot 10.5)}$$

$$= 106.237 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 365534 \cdot 0.1 / (10.5(228.0 + 106.24 + 106.24))$$

$$= -6.01 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -365534 / (4 \cdot 10.5(228.0 + 106.237 + 106.237)) -$$

$$12 \cdot 0.016 \cdot 365534 \cdot 1766.75 / (10560.0 \cdot 10.5^2)$$

$$= -127.62 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.0 + 1.56 \cdot \sqrt{(1766.75 \cdot 10.5)}, 2 \cdot 950.0)$$

$$= 440.47 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3080.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	508.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:



$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1761.5 \cdot 10.5)} + 300.0)^3) / 12$$

$$= 49407.250 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 49407.25 + 5377.665 (229.019 - 5.25)^2$$

$$= 269323552.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	512.2	10.5	5.2	5377.7	28232.7	0.269E+09
Wearplate	300.0	10.0	15.5	3000.0	46500.0	0.137E+09
Web	16.0	498.0	269.5	7968.0	2147378.8	0.178E+09
BasePlate	254.0	20.0	528.5	5080.0	2684781.8	0.456E+09
Totals	...	...	...	21425.7	4906893.0	0.104E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4906893 / 21425.668$$

$$= 229.019 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 365533.844$$

$$= 74394.0703 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.6361 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2300.0 - 1761.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 843.251 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 62758340.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13.8204 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(365534 + 8268) \cdot 254.0) / (4 \cdot 3080.0 \cdot 137.997)}$$

$$= 12.944 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (1760.0 + 12.0 + 10.0)$$

$$= 3086.514 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3086.5142 / (4 - 1)$$

$$= 1028.838 \text{ mm}$$

Baseplate Pressure Area [Ap]:



$$\begin{aligned}
 &= e \cdot Bpwid / 2 \\
 &= 1028.838 \cdot 254.0 / 2 \\
 &= 130662.430 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\
 &= 373802.094 / (3080.0 \cdot 254.0) \\
 &= 0.478 \text{ N/mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 130662.4 \cdot 0.48 \\
 &= 62432.117 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 3392.0 + 8230.704 \\
 &= 11622.704 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Compressive Stress [Sc]:} \\
 &= P / Ar \\
 &= 62432.1 / 11622.7041 \\
 &= 5.372 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	228.0	...	3648.0	...	0.158E+08

$$\begin{aligned}
 \text{Rib dimension [D]:} \\
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 228.0 - 16.0 \\
 &= 212.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance to Centroid from Datum [ytot]:} \\
 &= AY / A \\
 &= 0.0 / 11622.704 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance to Centroid [C1]:} \\
 &= \text{Saddle Width} / 2 \\
 &= 228.0 / 2 \\
 &= 114.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Radius of Gyration [r]:} \\
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(15803123 / 11622.704)} \\
 &= 36.874 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Length of Outer Rib, step 1, [Lv]:} \\
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2300.0 - \cos(120.0 / 2) (1760.0 + 12.0 + 10.0) - 20.0 \\
 &= 1389.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Length of Outer Rib, step 2, [Lw]:} \\
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120.0 / 2) (1760.0 + 12.0 + 10.0) \\
 &= 3086.514 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Length of Outer Rib, step 3, [Bd]:} \\
 &= \text{Baseplate length} - \text{clearance} \\
 &= 3080.0 - 50.8 \\
 &= 3029.200 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Length of Outer Rib, step 4, [Dd]:} \\
 &= (Lw - Bd) / 2
 \end{aligned}$$



$$= (3086.514 - 3029.2) / 2$$

$$= 28.657 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1389.0^2 + 28.657^2}$$

$$= 1389.296 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 202713}{207.0}}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1389.296 / 36.874$$

$$= 37.677$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 52433.5 / (2 \cdot 3080.0) \cdot 1028.838 \cdot 1389.3 / 2$$

$$= 6085770.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($37.677 < 139.0351$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (37.68)^2 / (2 \cdot 139.04^2)) 207 /$$

$$(5/3 + 3 \cdot (37.68) / (8 \cdot 139.04) - (37.68)^3 / (8 \cdot 139.04^3))$$

$$= 112.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.37 / 112.92 + (6085770 \cdot 114.0 / 15803138) / 138.0$$

$$= 0.366$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	212.0	0.0	3392.0	0.0	0.158E+08
Web	1028.8	16.0	0.0	16461.4	0.0	0.351E+06
Totals	...	...	...	19853.4	...	0.161E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 19853.408$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.0 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 2300.0 - \sqrt{(1782.0 + 10.0)^2 - (1028.838/2)^2} - 20.0$$

$$= 573.865 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{16148837 / 19853.408}$$

$$= 28.520 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 573.865 / 28.52$$

$$= 20.121$$

Unit Force [Force,u]:



$$\begin{aligned}
 &= F / (2 \cdot \text{Baseplate Length}) \\
 &= 52433.457 / (2 \cdot 3080.0) \\
 &= 8.512 \text{ N/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 8.512 \cdot 1028.838 \cdot 573.865 \\
 &= 5027598.000 \text{ N-mm}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 5027598 / 141656.578 \\
 &= 35.480 \text{ MPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($20.1213 < 139.0351$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (20.12)^2 / (2 \cdot 139.04^2)) 207 / \\
 &\quad (5/3 + 3 \cdot (20.12) / (8 \cdot 139.04) - (20.12^3) / (8 \cdot 139.04^3)) \\
 &= 119 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 6.29 / 119.05 + (5027598 \cdot 114.0 / 16148837) / 138.0 \\
 &= 0.310
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: N11-2''****From: 10**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

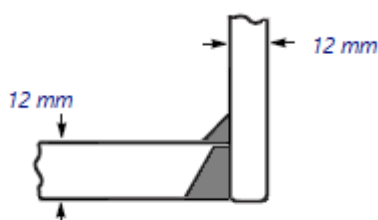
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N11-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 0.45)$
= 5.1895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N11-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7854 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2642 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2642 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.2642, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.7854, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.189, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.494$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N11-2''

Intermediate Calc. for nozzle/shell Welds T_{min} 10.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1015 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 212.1016 mm

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**Input, Nozzle Desc: N10A-2"****From: 10**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

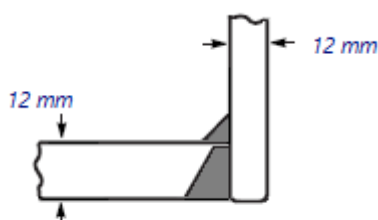
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	182.4740	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N10A-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 0.45)$
 $= 5.1895 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
 $= 0.0971 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2748 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	73.5268 mm
Parallel to Vessel Wall	Rn+tn+t	36.7634 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N10A-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 12.0, tr = 5.189, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.494$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N10A-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N10A-2''



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 0.45)$
= 5.1895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2748 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N10A-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7748 mm
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.2642 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.2642 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.7917 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm
Wall Thickness per table UG-45	tb3	= 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2642, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.7748, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Weld Size Calculations, Description: N10A-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000 mm
---	------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.5404 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 203.6541 mm

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**Input, Nozzle Desc: N10B-2"****From: 10**

Pressure for Reinforcement Calculations	P	0.4528	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

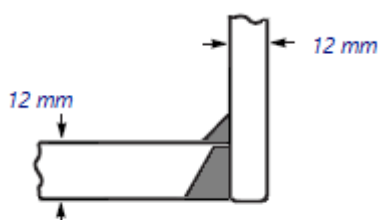
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	182.4740	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N10B-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
 $= 5.1996 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
 $= 0.0973 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2748 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	73.5268 mm
Parallel to Vessel Wall	Rn+tn+t	36.7634 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N10B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 12.0, tr = 5.2, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.495$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N10B-2''

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N10B-2''



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 0.45)$
= 5.1996 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0973 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2748 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N10B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7748 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2755 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2755 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2755, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.7748, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Weld Size Calculations, Description: N10B-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.5404 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 203.6541 mm

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**Input, Nozzle Desc: N3-24''****From: 20**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	5293.3325	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2550.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.98	MPa
Allowable Stress At Ambient	Sna	137.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	
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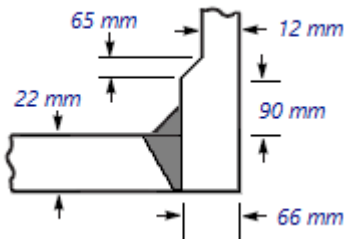
Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	65.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		66.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
 $= 5.7802 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.45 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.45)$
 $= 0.9970 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.2491 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 54.0 / 0.5773$
 $= 93.5308 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

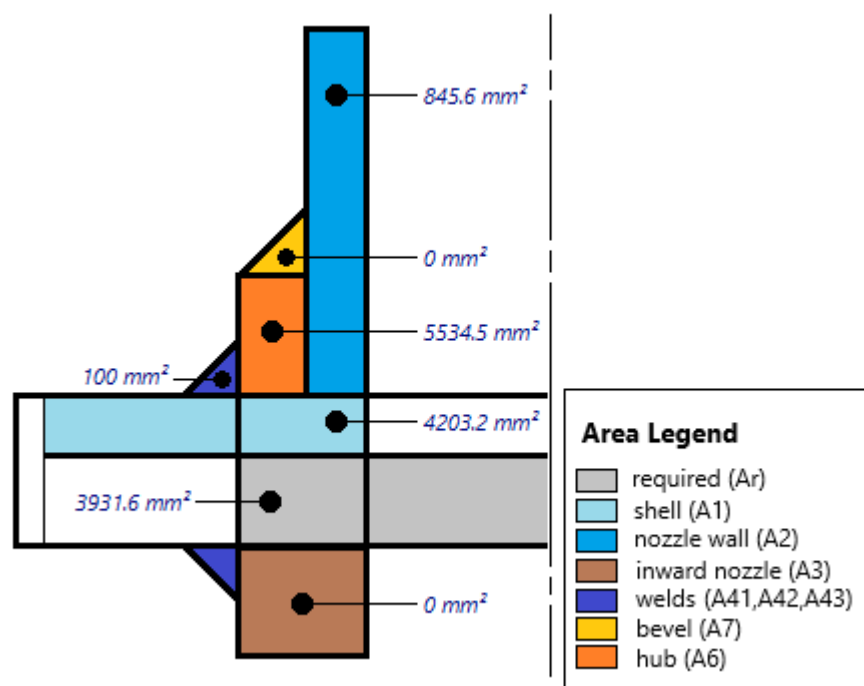
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1177.2001	mm
Parallel to Vessel Wall, opening length	d	588.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3402.319	3931.605	NA
Area in Shell	A1	8663.870	4203.166	NA
Area in Nozzle Wall	A2	973.964	845.638	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.990	99.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5534.472	5534.472	NA
TOTAL AREA AVAILABLE	Atot	15272.296	10683.265	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(588.6 \cdot 13.3589 \cdot 1 + 2 \cdot 64.5 \cdot 13.3589 \cdot 1(1 - 1.))$$

$$= 3931.605 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 588.6(1.0 \cdot 20.5 - 1.0 \cdot 13.359) - 2 \cdot 64.5$$

$$(1.0 \cdot 20.5 - 1.0 \cdot 13.3589) \cdot (1 - 1.)$$

$$= 4203.166 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 51.25)(10.5 - 2.25)0.9999$$

$$= 845.638 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9999 + (0.0)^2 \cdot 0.9999$$

$$= 99.990 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(51.2, 872.0, 90.0)) \cdot (66.0 - 12.0) \cdot 1.$$

$$= 5534.472 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7491 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(7.2802, 3.0)]$$

$$= 7.2802 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7491, 7.2802)$$

$$= 7.2802 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.3,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.7 MPa	Passed
Occasional	: 6.2,	Allowable	: 183.5 MPa	Passed
Shear	: 17.0,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.0, tr = 0.997, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.095, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 5.78, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.282, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 0.45/1.96 = 0.231$$



Weld Size Calculations, Description: N3-24"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (3931.605 - 4203.1655 + 2 \cdot 64.5 \cdot 0.9999 \cdot (1.0 \cdot 20.5 - 13.3589)) \cdot 138) \\
 &= 89629.23 \text{ N}
 \end{aligned}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A_2 + A_4 - (W_i - \text{Can}) / 7.07)^2 \cdot f_{r2} \cdot S_v \\
 &= (6380.1089 + 99.9905 - 0.0 \cdot 1.) \cdot 138 \\
 &= 894161.56 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (6380.1089 + 0.0 + 99.9905 + (2644.2473)) \cdot 138 \\
 &= 1259030.00 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (6380.1089 + 0.0 + 99.9905 + 0.0 + (2644.2473)) \cdot 138 \\
 &= 1259030.00 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 647372. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 299.55) \cdot (66.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 5862325. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 1173308. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (647372 + 5862325) = 6509697 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (647372 + 0 + 1173308 + 0) = 1820680 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (647372 + 1173308 + 0) = 1820680 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:



Path 1-1 = 6509697 N , must exceed W = 89629 N or W1 = 894161 N
 Path 2-2 = 1820680 N , must exceed W = 89629 N or W2 = 1259030 N
 Path 3-3 = 1820680 N , must exceed W = 89629 N or W3 = 1259030 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 26.5939 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 920.5939 mm

Input Echo, WRC297 Item 1, Description: N3-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3520.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	717.600	mm
Nozzle Thickness	66.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	137.98 MPa
Nozzle Hot S.I. Allowable	SNmh	137.98 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	0.45	MPa
Radial Load	P	-38398.29	N
Circumferential Shear	Vc	28798.71	N
Longitudinal Shear	Vl	38398.29	N
Circumferential Moment	Mc	0.461E+08	N-mm
Longitudinal Moment	Ml	0.599E+08	N-mm
Torsional Moment	Mt	0.691E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 717.6 + 2 \cdot 1.65 \cdot \sqrt{(1771.75(22.0 - 1.5))}$$

$$= 1346.515 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3543.500 mm
Nozzle Outside Diameter	(d) =	717.600 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	64.500 mm



$$\begin{aligned} T / t &= 0.318 \\ d / t &= 11.126 \end{aligned}$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.662$$

Nr/P	=	0.060
Mr/P	=	0.116
M0/P	=	0.035
N0/P	=	0.077
MrD/Mc	=	0.272
NrDL/Mc	=	0.092
M0d/Mc	=	0.082
N0DL/Mc	=	0.085
MrD/M1	=	0.094
NrDL/M1	=	0.063
M0D/M1	=	0.027
N0DL/M1	=	0.094

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	38	39	38	39
Outplane Stress Summary	13	40	116	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-112	112	112	-112
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-36	61	213	-138

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	51	62	213	151

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-75	75	75	-75
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-42	69	134	-55

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-249	249	249	-249
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	39	38	39
Inplane Stress Summary	-156	216	371	-254

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	156	216	371	254

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	-9	-8	28	28

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-4	1	16	-7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	10	10	28	36
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-3	-3	22	22
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-17	15	29	-21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	17	18	29	44
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**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	39	38	39	38	39	38	39
Circ. Pl (SUS)	-11	-11	25	25	-9	-9	19	19
Circ. Q (SUS)	-13	13	51	-51	-186	186	312	-312
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-7	-7	17	17	-5	-5	19	19
Long. Q (SUS)	-49	49	175	-175	-56	56	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	38.0	39.0	38.0	39.0	38.0	39.0	38.0	39.0
Pm+Pl (SUS)	27.1	28.1	63.0	64.0	29.1	30.1	57.1	58.0
Pm+Pl+Q (Total)	52.0	62.2	211.1	152.1	157.1	216.1	369.1	254.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	138.00	Passed
Pm+Pl (SUS)	64.04	207.00	Passed
Pm+Pl+Q (TOTAL)	369.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-11	-11	25	25	-5	-5	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-3	3	11	-11	-16	16	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	9.0	9.0	27.0	27.0	3.0	3.0	21.0	21.0
Pm+Pl+Q (Total)	9.2	10.2	27.1	34.1	17.1	18.1	27.2	42.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	137.98	Passed

WARM HP FLARE KO DRUM

PV Elite 28

Licensee: SL User



FileName : A170RHP10HN001_WARM HP FLARE K.O. DRUM-7%

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Nozzle Calcs.: N3-24''

Nozl: 38 3:11pm Sep 17, 2026



HEXAGON

Pm+Pl (SUS)	27.00	206.98	Passed
Pm+Pl+Q (TOTAL)	42.05	413.95	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: M1-24****From: 20**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	5293.3325	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.98	MPa
Allowable Stress At Ambient	Sna	137.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	65.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm

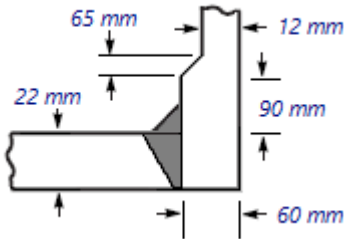
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
 $= 5.7802 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.45 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.45)$
 $= 0.9970 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.2491 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 48.0 / 0.5773$
 $= 83.1385 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

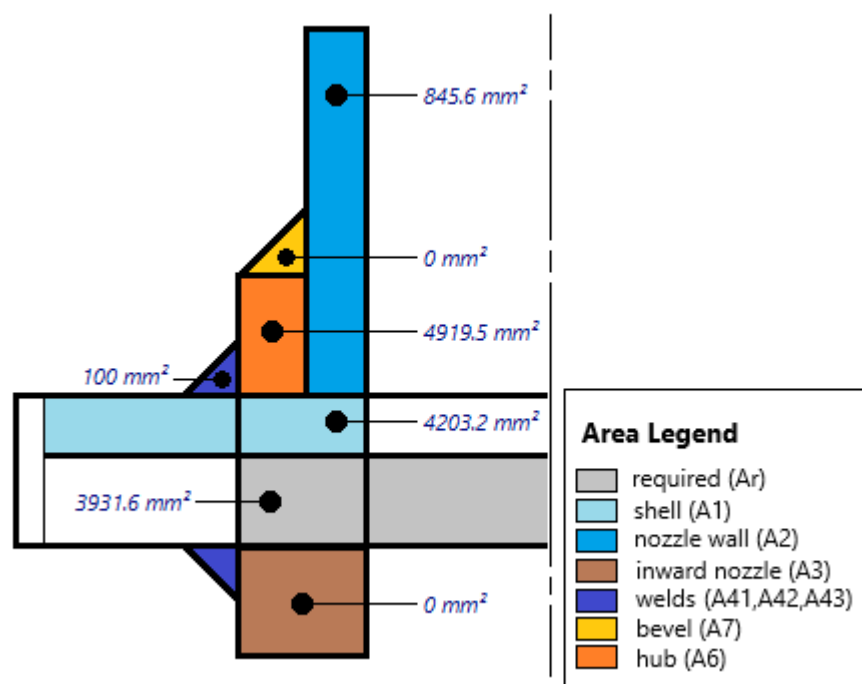
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1177.2001	mm
Parallel to Vessel Wall, opening length	d	588.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3402.313	3931.597	NA
Area in Shell	A1	8663.887	4203.174	NA
Area in Nozzle Wall	A2	973.964	845.638	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.990	99.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4919.531	4919.531	NA
TOTAL AREA AVAILABLE	Atot	14657.372	10068.333	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr_1)) \text{ per UG-37(d)}$$

$$= 0.5(588.6 \cdot 13.3589 \cdot 1 + 2 \cdot 58.5 \cdot 13.3589 \cdot 1(1 - 1.))$$

$$= 3931.597 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E_1 \cdot t - F \cdot tr) - 2 \cdot tn(E_1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$$

$$= 588.6(1.0 \cdot 20.5 - 1.0 \cdot 13.359) - 2 \cdot 58.5$$

$$(1.0 \cdot 20.5 - 1.0 \cdot 13.3589) \cdot (1 - 1.)$$

$$= 4203.174 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{lnp})(tn - trn)fr_2$$

$$= (2 \cdot 51.25)(10.5 - 2.25)0.9999$$

$$= 845.638 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 0.9999 + (0.0)^2 \cdot 0.9999$$

$$= 99.990 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(51.2, 872.0, 90.0)) \cdot (60.0 - 12.0) \cdot 1.$$

$$= 4919.531 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.7491 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.7491 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 12.0$, $t_r = 0.997$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.095$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 5.78$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.282$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 0.45/1.96 = 0.231$$

Weld Size Calculations, Description: M1-24

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (3931.5974 - 4203.1738 + 2 \cdot 58.5 \cdot 0.9999 \cdot (1.0 \cdot 20.5 - 13.3589)) \cdot 138)$$

$$= 77803.70 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$



$$= (5765.1685 + 99.9905 - 0.0 \cdot 1.) \cdot 138$$

$$= 809308.44 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (5765.1685 + 0.0 + 99.9905 + (2398.2712)) \cdot 138$$

$$= 1140235.62 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (5765.1685 + 0.0 + 99.9905 + 0.0 + (2398.2712)) \cdot 138$$

$$= 1140235.62 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 647372. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 299.55) \cdot (60.0 - 1.5) \cdot 0.7 \cdot 138$$

$$= 5316992. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1173308. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (647372 + 5316993) = 5964365 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (647372 + 0 + 1173308 + 0) = 1820680 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (647372 + 1173308 + 0) = 1820680 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 5964364 N	, must exceed W = 77803 N	or W1 = 809308 N
Path 2-2 = 1820680 N	, must exceed W = 77803 N	or W2 = 1140235 N
Path 3-3 = 1820680 N	, must exceed W = 77803 N	or W3 = 1140235 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 26.5939 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 920.5939 mm

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**Input, Nozzle Desc: N4-24''****From: 30**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	5293.3325	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4050.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.98	MPa
Allowable Stress At Ambient	Sna	137.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	
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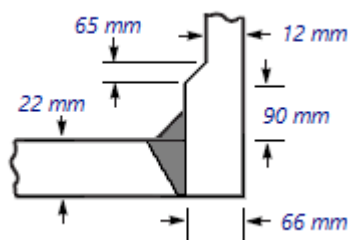
Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	65.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		66.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
 $= 5.7802 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.45 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.45)$
 $= 0.9970 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.2491 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 54.0 / 0.5773$
 $= 93.5308 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1177.2001	mm
Parallel to Vessel Wall, opening length	d	588.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	51.2500	mm

Weld Strength Reduction Factor [fr1]:

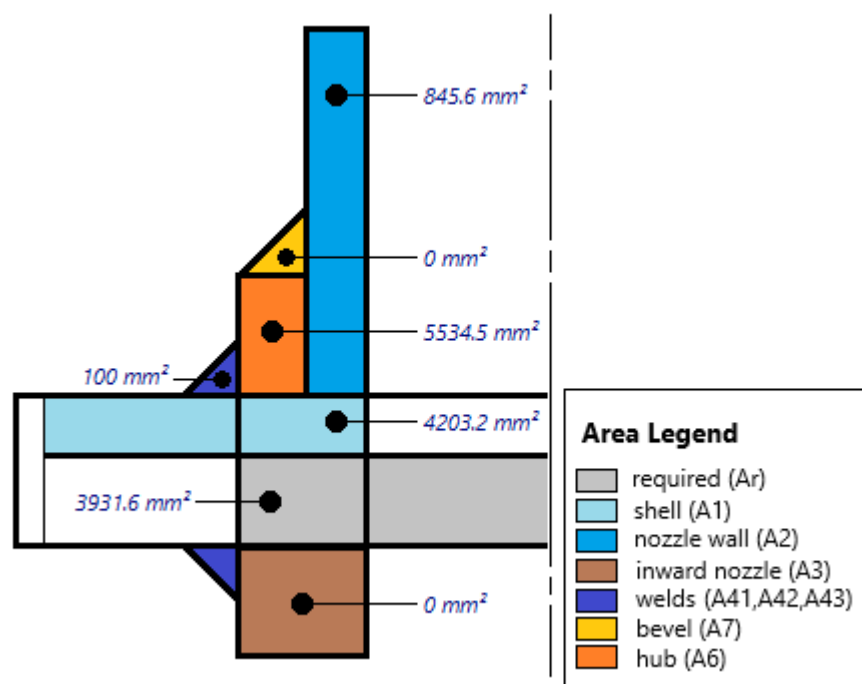
$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3402.319	3931.605	NA
Area in Shell	A1	8663.870	4203.166	NA
Area in Nozzle Wall	A2	973.964	845.638	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.990	99.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5534.472	5534.472	NA
TOTAL AREA AVAILABLE	Atot	15272.296	10683.265	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr_1)) \text{ per UG-37(d)}$$

$$= 0.5(588.6 \cdot 13.3589 \cdot 1 + 2 \cdot 64.5 \cdot 13.3589 \cdot 1(1 - 1.))$$

$$= 3931.605 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E_1 \cdot t - F \cdot tr) - 2 \cdot tn(E_1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$$

$$= 588.6(1.0 \cdot 20.5 - 1.0 \cdot 13.359) - 2 \cdot 64.5$$

$$(1.0 \cdot 20.5 - 1.0 \cdot 13.3589) \cdot (1 - 1.)$$

$$= 4203.166 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{lnp})(tn - trn)fr_2$$

$$= (2 \cdot 51.25)(10.5 - 2.25)0.9999$$

$$= 845.638 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 0.9999 + (0.0)^2 \cdot 0.9999$$

$$= 99.990 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(51.2, 872.0, 90.0)) \cdot (66.0 - 12.0) \cdot 1.0$$

$$= 5534.472 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7491 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.8312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.831, \max(7.2802, 3.0)]$$

$$= 7.2802 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7491, 7.2802)$$

$$= 7.2802 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.3,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.7 MPa	Passed
Occasional	: 6.2,	Allowable	: 183.5 MPa	Passed
Shear	: 17.0,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.0, tr = 0.997, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.095, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 5.78, c = 1.5 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.282, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.45/1.96 = 0.231



Weld Size Calculations, Description: N4-24"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (3931.605 - 4203.1655 + 2 \cdot 64.5 \cdot 0.9999 \cdot \\
 &\quad (1.0 \cdot 20.5 - 13.3589)) \cdot 138) \\
 &= 89629.23 \text{ N}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can}) / 7.07)^2 \cdot f_{r2} \cdot S_v \\
 &= (6380.1089 + 99.9905 - 0.0 \cdot 1.) \cdot 138 \\
 &= 894161.56 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (6380.1089 + 0.0 + 99.9905 + (2644.2473)) \cdot 138 \\
 &= 1259030.00 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (6380.1089 + 0.0 + 99.9905 + 0.0 + (2644.2473)) \cdot 138 \\
 &= 1259030.00 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 647372. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 299.55) \cdot (66.0 - 1.5) \cdot 0.7 \cdot 138 \\
 &= 5862325. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 1173308. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (647372 + 5862325) = 6509697 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (647372 + 0 + 1173308 + 0) = 1820680 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (647372 + 1173308 + 0) = 1820680 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:



Path 1-1 = 6509697 N , must exceed W = 89629 N or W1 = 894161 N
 Path 2-2 = 1820680 N , must exceed W = 89629 N or W2 = 1259030 N
 Path 3-3 = 1820680 N , must exceed W = 89629 N or W3 = 1259030 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 26.5939 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 920.5939 mm

Input Echo, WRC297 Item 1, Description: N4-24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3520.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	717.600	mm
Nozzle Thickness	66.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	137.98 MPa
Nozzle Hot S.I. Allowable	SNmh	137.98 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	0.45	MPa
Radial Load	P	-38398.29	N
Circumferential Shear	Vc	28798.71	N
Longitudinal Shear	Vl	38398.29	N
Circumferential Moment	Mc	0.461E+08	N-mm
Longitudinal Moment	Ml	0.599E+08	N-mm
Torsional Moment	Mt	0.691E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 717.6 + 2 \cdot 1.65 \cdot \sqrt{(1771.75(22.0 - 1.5))}$$

$$= 1346.515 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3543.500 mm
Nozzle Outside Diameter	(d) =	717.600 mm
Vessel Thickness used	(T) =	20.500 mm
Nozzle Thickness used	(t) =	64.500 mm



$$\begin{aligned} T / t &= 0.318 \\ d / t &= 11.126 \end{aligned}$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.662$$

Nr/P	=	0.060
Mr/P	=	0.116
M0/P	=	0.035
N0/P	=	0.077
MrD/Mc	=	0.272
NrDL/Mc	=	0.092
M0d/Mc	=	0.082
N0DL/Mc	=	0.085
MrD/M1	=	0.094
NrDL/M1	=	0.063
M0D/M1	=	0.027
N0DL/M1	=	0.094

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	38	39	38	39
Outplane Stress Summary	13	40	116	13

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-112	112	112	-112
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-36	61	213	-138

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	51	62	213	151

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-75	75	75	-75
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19

Outplane Stress Summary	-42	69	134	-55
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-249	249	249	-249
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	39	38	39

Inplane Stress Summary	-156	216	371	-254
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	156	216	371	254
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-9	-8	28	28
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-4	1	16	-7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	10	10	28	36
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-3	-3	22	22
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-17	15	29	-21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	17	18	29	44
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**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	39	38	39	38	39	38	39
Circ. Pl (SUS)	-11	-11	25	25	-9	-9	19	19
Circ. Q (SUS)	-13	13	51	-51	-186	186	312	-312
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-7	-7	17	17	-5	-5	19	19
Long. Q (SUS)	-49	49	175	-175	-56	56	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	38.0	39.0	38.0	39.0	38.0	39.0	38.0	39.0
Pm+Pl (SUS)	27.1	28.1	63.0	64.0	29.1	30.1	57.1	58.0
Pm+Pl+Q (Total)	52.0	62.2	211.1	152.1	157.1	216.1	369.1	254.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	138.00	Passed
Pm+Pl (SUS)	64.04	207.00	Passed
Pm+Pl+Q (TOTAL)	369.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-11	-11	25	25	-5	-5	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-3	3	11	-11	-16	16	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	9.0	9.0	27.0	27.0	3.0	3.0	21.0	21.0
Pm+Pl+Q (Total)	9.2	10.2	27.1	34.1	17.1	18.1	27.2	42.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	137.98	Passed

WARM HP FLARE KO DRUM

PV Elite 28

Licensee: SL User



FileName : A170RHP10HN001_WARM HP FLARE K.O. DRUM-7%

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Nozzle Calcs.: N4-24''

Nozl: 40 3:11pm Sep 17,2026



HEXAGON

Pm+Pl (SUS)	27.00	206.98	Passed
Pm+Pl+Q (TOTAL)	42.05	413.95	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: N5-2''****From: 30**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	2000.0004	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5450.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

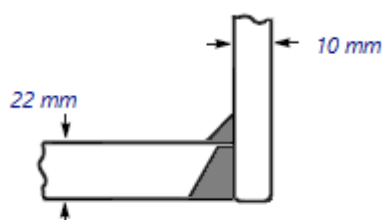
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N5-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
= 5.7802 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P / (S_n \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4425 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	91.8000 mm
Parallel to Vessel Wall	Rn+tn+t	45.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N5-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9425 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.2802, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.9425, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N5-2''

Intermediate Calc. for nozzle/shell Welds T_{min} 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1833 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 894.1833 mm

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**Input, Nozzle Desc: N6-2''****From: 40**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6250.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

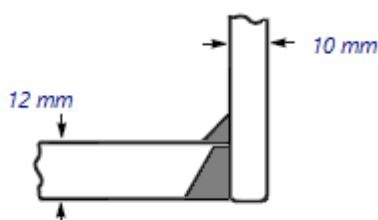
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	888.0001	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N6-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
= 5.7802 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P/(S_n \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4427 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N6-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9427 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.2802, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9427, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1833 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 900.1834 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 35.90 mm

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**Input, Nozzle Desc: N9-3''****From: 40**

Pressure for Reinforcement Calculations	P	0.4603	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	1999.9996	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7250.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	
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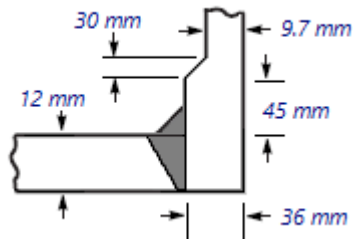
Hub Height of Integral Nozzle	h	45.0000	mm
Height of Beveled Transition	L'	30.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		36.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N9-3"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.46 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.46)$
 $= 5.8870 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.46 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 0.46)$
 $= 0.1731 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3956 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 26.265 / 0.5773$
 $= 45.4931 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.8618 mm
Parallel to Vessel Wall, opening length	d	72.4309 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N9-3".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8956 mm
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.3870 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3870 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm
Wall Thickness per table UG-45	tb3	= 6.3000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(7.387, 3.0)]$
 $= 6.3000 \text{ mm}$



Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.8956, 6.3)

= 6.3000 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	265.7 MPa	Passed
Occasional	:	0.9,	Allowable	:	156.9 MPa	Passed
Shear	:	22.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.173, c = 1.5 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.021, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.887, c = 1.5 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.561, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 0.46/5.10 = 0.090

Weld Size Calculations, Description: N9-3"

Intermediate Calc. for nozzle/shell Welds Tmin 10.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5614 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.5614 mm

Input Echo, WRC297 Item 1, Description: N9-3" :



Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3520.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	141.431	mm
Nozzle Thickness	36.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	117.98 MPa
Nozzle Hot S.I. Allowable	SNmh	117.98 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.46	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (3.929) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 141.431 + 2 \cdot 1.65 \cdot \sqrt{(1766.75(12.0 - 1.5))}$$

$$= 590.896 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3533.500	mm
Nozzle Outside Diameter	(d) =	141.431	mm
Vessel Thickness used	(T) =	10.500	mm
Nozzle Thickness used	(t) =	34.500	mm
T / t	=	0.304	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.734$$

Nr/P	=	0.123
Mr/P	=	0.270
M0/P	=	0.079
N0/P	=	0.140
MrD/Mc	=	0.313
NrDL/Mc	=	0.067



$M\theta d/Mc = 0.097$
 $N\theta DL/Mc = 0.027$
 $MrD/Ml = 0.235$
 $NrDL/Ml = 0.087$
 $M\theta D/Ml = 0.074$
 $N\theta DL/Ml = 0.091$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-26	26	26	-26
Normal Pressure Stress	76	77	76	77
Outplane Stress Summary	71	84	135	41

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	24	52	204	-106

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	72	84	204	147

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-26	26	26	-26
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	37	49	93	-1

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-86	86	86	-86
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	76	77	76	77
Inplane Stress Summary	63	95	242	-71

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	63	95	243	72

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	1	1	11	12

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-6	6	6	-6
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-4	-1	15	-7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0



Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	4	15	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	5	5	7	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-7	7	7	-7
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-4	0	15	-8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	9	5	15	17

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	77	76	77	76	77	76	77
Circ. Pl (SUS)	1	1	11	11	2	2	8	8
Circ. Q (SUS)	-6	6	46	-46	-16	16	156	-156



Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	0	0	10	10	5	5	7	7
Long. Q (SUS)	-14	14	154	-154	-6	6	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	76.0	77.0	76.0	77.0	76.0	77.0	76.0	77.0
Pm+Pl (SUS)	77.0	78.0	87.0	88.0	78.1	79.1	84.1	85.1
Pm+Pl+Q (Total)	71.3	84.5	202.1	148.1	62.0	95.0	240.2	71.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	138.00	Passed
Pm+Pl (SUS)	88.02	207.00	Passed
Pm+Pl+Q (TOTAL)	240.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	1	1	11	11	5	5	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-1	1	11	-11	-2	2	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	4.0	4.0	11.0	11.0	7.0	7.0	7.0	7.0
Pm+Pl+Q (Total)	5.0	3.0	14.0	19.0	9.0	5.0	14.0	17.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	117.98	Passed
Pm+Pl (SUS)	11.00	176.97	Passed
Pm+Pl+Q (TOTAL)	19.00	353.93	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**Input, Nozzle Desc: N1-2''****From: 40**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6800.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

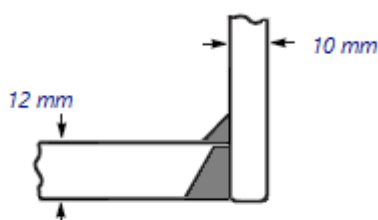
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
= 5.7802 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P / (S_n \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4425 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9425 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2802, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.9425, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N1-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1833 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 884.1833 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 35.90 mm

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**Input, Nozzle Desc: N2-2''****From: 40**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	1999.9996	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7500.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

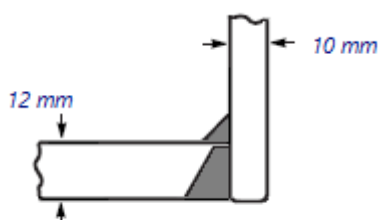
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	871.9999	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
= 5.7802 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P / (S_n \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4425 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N2-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9425 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2802, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.9425, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1833 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 884.1833 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 35.90 mm

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**Input, Nozzle Desc: N7-2''****From: 50**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3520.00	mm
Design Length of Section	L	1853.3335	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9110.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

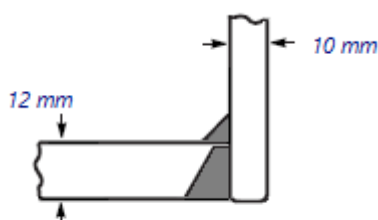
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	888.0001	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N7-2''**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.45 \cdot 1761.4999 / (138 \cdot 1.0 - 0.6 \cdot 0.45)$
= 5.7802 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P / (S_n \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4427 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	71.8000 mm
Parallel to Vessel Wall	Rn+tn+t	35.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N7-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9427 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2802 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7936 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.2802, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9427, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	48.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	246.7 MPa	Passed
Occasional	:	0.4,	Allowable	:	156.9 MPa	Passed
Shear	:	40.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 0.45/5.10 = 0.089

Weld Size Calculations, Description: N7-2"

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1833 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 900.1834 mm

Input Echo, WRC297 Item 1, Description: N7-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3520.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	10.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	117.98 MPa
Nozzle Hot S.I. Allowable	SNmh	117.98 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.45	MPa
Radial Load	P	-3199.86	N
Circumferential Shear	Vc	2399.89	N
Longitudinal Shear	Vl	3199.86	N
Circumferential Moment	Mc	320112.56	N-mm
Longitudinal Moment	Ml	416146.78	N-mm
Torsional Moment	Mt	480169.53	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Warning - The ratio, D_n/T_v , (4.233) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (5.080) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 50.8 + 2 \cdot 1.65 \cdot \sqrt{(1766.75(12.0 - 1.5))}$$

$$= 500.265 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3533.500 mm
Nozzle Outside Diameter	(d) =	50.800 mm
Vessel Thickness used	(T) =	10.500 mm
Nozzle Thickness used	(t) =	8.500 mm
T / t	=	1.235
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.264$$

Nr/P	=	0.231
Mr/P	=	0.345
M0/P	=	0.141
N0/P	=	0.198
MrD/Mc	=	0.280
NrDL/Mc	=	0.155
M0d/Mc	=	0.135
N0DL/Mc	=	-0.126
MrD/Ml	=	0.273
NrDL/Ml	=	0.158
M0D/Ml	=	0.130
N0DL/Ml	=	-0.075

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	5	5	-5	-5
Outplane Bending (ML)	-58	58	58	-58
Normal Pressure Stress	75	76	75	76
Outplane Stress Summary	53	120	158	-6

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-121	121	121	-121
Inplane Pressure Stress	37	37	37	37
Inplane Stress Summary	-28	94	237	-125

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	11	11	11	11

Shear Stress Summary	14	14	8	8
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	86	127	238	125
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-46	46	46	-46
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	37	37	37	37

Outplane Stress Summary	28	72	107	-34
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-96	96	96	-96
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	75	76	75	76

Inplane Stress Summary	37	109	247	-64
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	11	11	11	11

Shear Stress Summary	7	7	15	15
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	41	111	248	70
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	5	5	-5	-5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1

Hoop Stress Summary	12	12	1	1
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-40	-40	40	40
Axial Bending (ML)	-131	131	131	-131
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-107	33	236	-149
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13

Shear Stress Summary	17	17	10	10
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	124	43	236	152

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	1	0	1

Hoop Stress Summary	13	14	0	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	60	-60	60	-60
Axial Membrane (Mc)	-30	-30	30	30
Axial Bending (Mc)	-106	106	106	-106
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0



Axial Stress Summary -72 17 201 -132

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	13	13	13	13

Shear Stress Summary 9 9 18 18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	88	25	205	138

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	75	76	75	76	75	76	75	76
Circ. Pl (SUS)	10	10	0	0	-2	-2	14	14
Circ. Q (SUS)	-34	34	82	-82	-36	36	156	-156
Long. Pm (SUS)	37	37	37	37	37	37	37	37
Long. Pl (SUS)	-5	-5	17	17	12	12	-2	-2
Long. Q (SUS)	-61	61	181	-181	-22	22	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	75.0	76.0	75.0	76.0	75.0	76.0	75.0	76.0
Pm+Pl (SUS)	85.1	86.1	75.2	76.2	73.4	74.4	89.2	90.2
Pm+Pl+Q (Total)	84.1	125.2	236.0	127.7	41.4	111.6	246.4	71.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	138.00	Passed
Pm+Pl (SUS)	90.16	207.00	Passed
Pm+Pl+Q (TOTAL)	246.39	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	1	0	1	0	1	0	1
Circ. Pl (SUS)	10	10	0	0	12	12	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-38	-38	42	42	-28	-28	32	32
Long. Q (SUS)	-71	71	191	-191	-46	46	166	-166
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0
Pm+Pl (SUS)	48.4	49.4	42.4	42.2	40.8	41.8	34.9	34.0
Pm+Pl+Q (Total)	123.2	41.4	233.9	151.3	87.9	24.8	202.9	137.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.00	117.98	Passed
Pm+Pl (SUS)	49.37	176.97	Passed
Pm+Pl+Q (TOTAL)	233.86	353.93	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: M2-24''****From: 60**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.98	MPa
Allowable Stress At Ambient	Sna	137.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	90.0000	mm
Height of Beveled Transition	L'	65.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		48.0000	mm

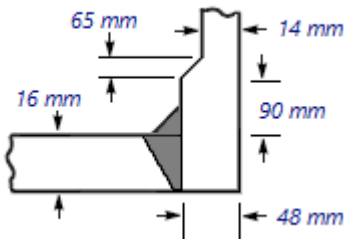
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
 $= 5.1895 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.45 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 0.45)$
 $= 0.9970 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4451 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34.0 / 0.5773$
 $= 58.8898 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

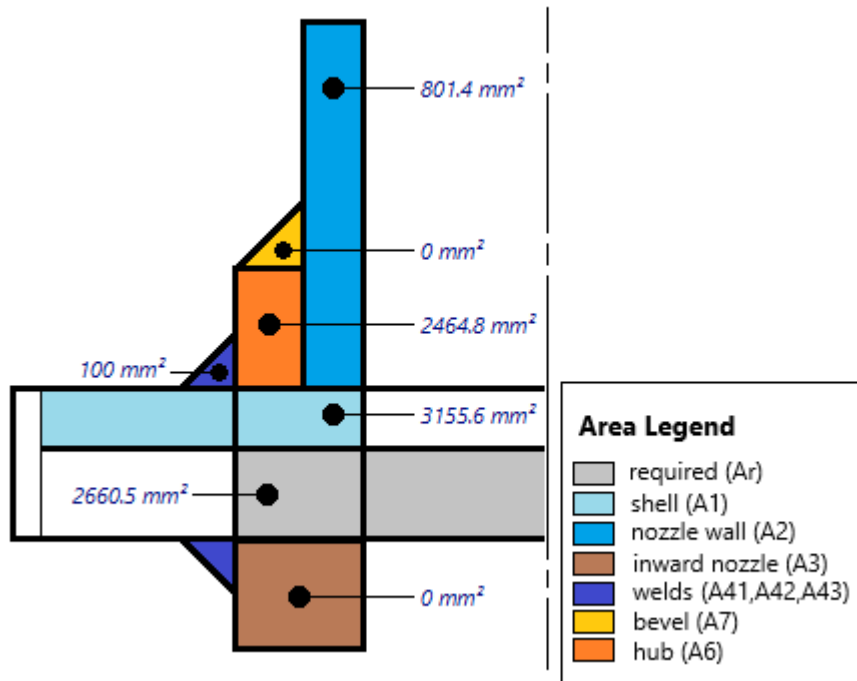
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	1169.2001	mm
Parallel to Vessel Wall, opening length	d	584.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3033.811	2660.546	NA
Area in Shell	A1	5442.853	3155.640	NA
Area in Nozzle Wall	A2	833.888	801.402	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.990	99.990	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	2464.765	2464.765	NA
TOTAL AREA AVAILABLE	Atot	8841.496	6521.797	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(584.6 \cdot 9.102 \cdot 1 + 2 \cdot 46.5 \cdot 9.102 \cdot 1(1 - 1.))$$

$$= 2660.546 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 584.6(1.0 \cdot 14.5 - 1.0 \cdot 9.102) - 2 \cdot 46.5$$

$$(1.0 \cdot 14.5 - 1.0 \cdot 9.102) \cdot (1 - 1.)$$

$$= 3155.640 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 36.25)(12.5 - 1.45) \cdot 0.9999$$

$$= 801.402 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9999 + (0.0)^2 \cdot 0.9999$$

$$= 99.990 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp,ho}, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(36.2, 300.0, 90.0)) \cdot (48.0 - 14.0) \cdot 1.0$$

$$= 2464.765 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.9451 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.9451 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 14.0$, $t_r = 0.997$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.08$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, $t_g = 16.0$, $t_r = 5.189$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.358$, Temp. Reduction = 72 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (2660.5459 - 3155.6401 + 2 \cdot 46.5 \cdot 0.9999 \cdot (1.0 \cdot 14.5 - 9.102)) \cdot 138)$$

$$= 948.62 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot S_v$$

$$= (3266.1663 + 99.9905 - 0.0 \cdot 1.0) \cdot 138$$

$$= 464481.78 \text{ N}$$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (3266.1663 + 0.0 + 99.9905 + (1348.3713)) \cdot 138$$

$$= 650537.88 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (3266.1663 + 0.0 + 99.9905 + 0.0 + (1348.3713)) \cdot 138$$

$$= 650537.88 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 647372. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 298.55) \cdot (48.0 - 1.5) \cdot 0.7 \cdot 138$$

$$= 4212218. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1368860. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (647372 + 4212218) = 4859590 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (647372 + 0 + 1368860 + 0) = 2016231 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (647372 + 1368860 + 0) = 2016231 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4859589 N	, must exceed W = 948 N	or W1 = 464481 N
Path 2-2 = 2016231 N	, must exceed W = 948 N	or W2 = 650537 N
Path 3-3 = 2016231 N	, must exceed W = 948 N	or W3 = 650537 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 14.6375 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 330.6375 mm

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**Input, Nozzle Desc: N8A-2''****From: 60**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

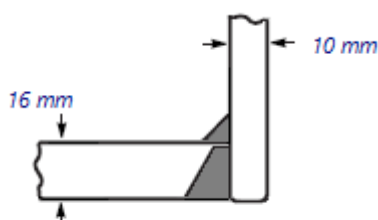
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N8A-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
 $= 5.1895 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P/(Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
 $= 0.0971 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	81.7496 mm
Parallel to Vessel Wall	Rn+tn+t	40.8748 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N8A-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45/1.96 = 0.231$

Weld Size Calculations, Description: N8A-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.**Reinforcement CALCULATION, Description: N8A-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
= 5.1895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P/(S \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	79.8000	mm
Parallel to Vessel Wall	Rn+tn+t	39.9000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N8A-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7854	mm
Wall Thickness per UG16(b),	tr16b = 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2642	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2642	mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000	mm
Wall Thickness per table UG-45	tb3 = 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2642, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.7854, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: N8A-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.5309 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3808 mm

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**Input, Nozzle Desc: N8B-2''****From: 60**

Pressure for Reinforcement Calculations	P	0.4528	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

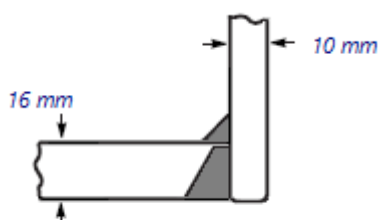
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N8B-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
 $= 5.1996 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P / (Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
 $= 0.0973 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	81.7496 mm
Parallel to Vessel Wall	Rn+tn+t	40.8748 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N8B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45 / 1.96 = 0.231$

Weld Size Calculations, Description: N8B-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.**Reinforcement CALCULATION, Description: N8B-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
= 5.1996 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P/(S \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0973 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	79.8000	mm
Parallel to Vessel Wall	Rn+tn+t	39.9000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N8B-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7854	mm
Wall Thickness per UG16(b),	tr16b = 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2755	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2755	mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000	mm
Wall Thickness per table UG-45	tb3 = 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.2755, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.7854, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: N8B-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.5309 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3808 mm

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**Input, Nozzle Desc: N12A****From: 60**

Pressure for Reinforcement Calculations	P	0.4519	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1300.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

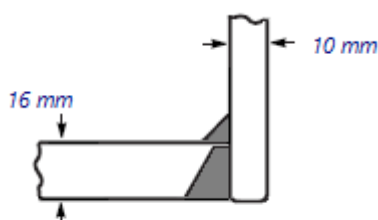
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N12A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
 $= 5.1895 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P / (Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.45 / (117.98 \cdot 1.0)))$
 $= 0.0971 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	84.5223 mm
Parallel to Vessel Wall	Rn+tn+t	42.2612 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N12A.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.097, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.011$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.45 / 1.96 = 0.231$

Weld Size Calculations, Description: N12A

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.**Reinforcement CALCULATION, Description: N12A**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.45 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.45)$
= 5.1895 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P/(S \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.45/(117.98 \cdot 1.0)))$
= 0.0971 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	79.8000 mm
Parallel to Vessel Wall	Rn+tn+t	39.9000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12A.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7854 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.2642 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.2642 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.2642, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.7854, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: N12A

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.9500 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.5082 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.0306 mm

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**Input, Nozzle Desc: N12B-2"****From: 60**

Pressure for Reinforcement Calculations	P	0.4558	MPa
Temperature for Internal Pressure	Temp	145	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	145	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	3520.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1300.0000	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.98	MPa
Allowable Stress At Ambient	Sna	117.98	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

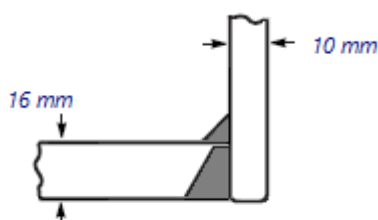
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: N12B-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.46 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.46)$
 $= 5.2334 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= Ro(1 - \exp(-P / (Sn \cdot E)))$ per Appendix 1-2 (a)(1)
 $= 25.4(1 - \exp(-0.46 / (117.98 \cdot 1.0)))$
 $= 0.0979 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	84.5223 mm
Parallel to Vessel Wall	Rn+tn+t	42.2612 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N12B-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 10.0, tr = 0.098, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.012$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.46 / 1.96 = 0.233$

Weld Size Calculations, Description: N12B-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.**Reinforcement CALCULATION, Description: N12B-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.46 \cdot 3523.0 \cdot 0.8993 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.46)$
= 5.2334 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R_o(1 - \exp(-P/(S \cdot E)))$ per Appendix 1-2 (a)(1)
= $25.4(1 - \exp(-0.46/(117.98 \cdot 1.0)))$
= 0.0979 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2854 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	79.8000	mm
Parallel to Vessel Wall	Rn+tn+t	39.9000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12B-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7854	mm
Wall Thickness per UG16(b),	tr16b = 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.3130	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.3130	mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7917	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000	mm
Wall Thickness per table UG-45	tb3 = 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(7.313, 3.0)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.7854, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Weld Size Calculations, Description: N12B-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 11.5082 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.0306 mm

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**Noted issues for the current model:**

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard Flange is limiting the MAWP which may affect the pressure used in the nozzle reinforcement calculations.**

Please review all reports carefully!

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: